Arc Reactor 기능 가이드 (통합 기능 카탈로그)

작성일: 2026-04-06
목적: Arc Reactor 전체 생태계 기능을 엔지니어링/운영 관점에서 정리

0) 자동 분석 블록 (2분 간격 동기화)

이 섹션은 2분마다 cron으로 자동 재생성됩니다.

- 마지막 동기화: 2026-04-06 08:24:21 +0900
- 분석 범위: /Users/jinan/ai 내 아슬란 생태계 프로젝트 8개 (arc-reactor, aslan-iam, swagger-mcp-server, atlassian-mcp-server, clipping-mcp-server, arc-reactor-admin, arc-reactor-web, Aslan-Verse-Web)
- 생성 방식: README/ECOSYSTEM.md 기반 정합성 스냅샷 + git 메타데이터

0) 작성 규칙(문체/근거/의사결정)

- 톤: 임원 보고용으로 한국어 존댓말(입니다/됩니다) 사용
- 증거 우선: 기능/관계/운영 포인트는 가능한 경우 구현 파일(경로·라인)로 뒷받침
- 미확인 항목 표기: "미확인(운영 연동/실행 검증 필요)"로 분리
- 반복 억제: 동일 주장 재사용을 피하고, 단락당 핵심 포인트 1개 이하 정렬
- 용어 정합성: Arc Reactor, MCP, IAM, ReAct, Slack, SSE를 일관된 표기 사용
- CTO 문서 반영 가이드: 각 섹션 말미에 '의사결정 포인트' 1개 이상 삽입(리스크/기대효과/우선순위)
- 독자 가이드:
  - 임원/CTO: 지금 문서는 "왜 이 조합이 사업/조직에 필요한가"에 대한 판단 근거를 빠르게 찾을 수 있게 정리합니다.
  - 기획/PO: 사용자 가치, 업무 적용 시나리오, 운영 제약을 중심으로 읽을 수 있게 핵심 동선을 앞부분에 배치합니다.
  - 개발자: API/도구/연동 코어 근거를 코드 라인 단위로 추적 가능한 형태로 정리합니다.
  - 운영/보안: 배포·가드·모니터링·권한 관리 포인트를 점검 항목 형태로 정리해 추적하기 쉽게 합니다.

1) 한눈에 보는 핵심 요약(직무 공통)

- 이 문서는 "무엇이 있는지"와 "왜 쓸만한지"를 별도로 구분해 보여줍니다.
- Arc Reactor는 aslan-iam 인증 기반 + MCP 생태계 동적 등록 + 웹/슬랙 채널 확장 + 관리 콘솔로 이어지는 중앙형 플랫폼입니다.
- 실무 도입 시점에는 기능 존재 여부보다 "운영 연동, 정책 일관성, 비용/안전 통제"의 확인이 우선입니다.

2) 프로젝트 상태 스냅샷

프로젝트	경로	역할	상태	근거 README	README 제목	브랜치	커밋	워크트리	갱신
arc-reactor	/Users/jinan/ai/arc-reactor	핵심 런타임	OK	/Users/jinan/ai/arc-reactor/README.ko.md	Arc Reactor	main	1773c962	dirty	77 minutes ago
aslan-iam	/Users/jinan/ai/aslan-iam	중앙 인증(IAM)	OK	/Users/jinan/ai/aslan-iam/README.md	aslan-iam	main	ee88eb3	dirty	3 days ago
swagger-mcp-server	/Users/jinan/ai/swagger-mcp-server	API 도구 MCP	OK	/Users/jinan/ai/swagger-mcp-server/README.md	Swagger MCP Server	main	8964374	clean	2 weeks ago
atlassian-mcp-server	/Users/jinan/ai/atlassian-mcp-server	Jira/Confluence/Bitbucket MCP	OK	/Users/jinan/ai/atlassian-mcp-server/README.md	atlassian-mcp-server	main	054a127	clean	2 weeks ago
clipping-mcp-server	/Users/jinan/ai/clipping-mcp-server	클리핑·요약 MCP	OK	/Users/jinan/ai/clipping-mcp-server/README.md	clipping-mcp-server	main	23abc52	clean	10 hours ago

프로젝트	경로	역할	상태	근거 README	README 제목	브랜치	커밋	워크트리	갱신
arc-reactor-admin	/Users/jinan/ai/arc-reactor-admin	운영/모니터링 UI	OK	/Users/jinan/ai/arc-reactor-admin/README.md	arc-reactor-admin	main	ca42355	clean	10 hours ago
arc-reactor/arc-web	/Users/jinan/ai/arc-reactor/arc-web	채팅/REST 모듈(내부)	OK	-	-	main	1773c962	dirty	77 minutes ago
arc-reactor/arc-slack	/Users/jinan/ai/arc-reactor/arc-slack	슬랙 채널 모듈(내부)	OK	-	-	main	1773c962	dirty	77 minutes ago
arc-reactor-web	/Users/jinan/ai/arc-reactor-web	사용자 UI	OK	/Users/jinan/ai/arc-reactor-web/README.ko.md	Arc Reactor Web	main	e0d84a8	clean	9 hours ago
Aslan-Verse-Web	/Users/jinan/ai/Aslan-Verse-Web	실험형 멀티 페르소나 조직 플랫폼	OK	-	-	main	c20bbdb	clean	20 hours ago

3) 아슬란 세계관 관점 관계(문서 근거 기반)

- Arc Reactor의 인증 중심은 aslan-iam의 공개키 검증 흐름과 결합되어 JWT를 로컬 검증하고, 로그인·권한·토큰 발급은 aslan-iam에서 분리 운영합니다 (`aslan-iam/README.md`, `aslan-iam/README.md`의 API 항목).
- Arc Reactor는 MCP 서버를 실행 중 등록만으로 붙이기 때문에, swagger/atlassian/clipping MCP는 `POST /api/mcp/servers` 중심의 런타임 등록/프록시 구조로 관리됩니다 (`arc-reactor/README.ko.md` 및 `arc-reactor/ECOSYSTEM.md`).
- 관리자 채널(`arc-reactor-admin`)은 arc-reactor Admin API를 통해 MCP preflight, 정책, 스케줄러, 도구 정책을 검증/운영하는 데 사용됩니다 (`arc-reactor-admin/README.md`, `arc-reactor/ECOSYSTEM.md`).
- 웹 채널(`arc-reactor-web`)은 세션 채팅, SSE 스트리밍, Persona/Tool Policy/Output Guard/스케줄러 관리 화면을 제공합니다 (`arc-reactor-web/README.ko.md`).
- Slack 채널(`arc-slack`)은 Socket Mode 또는 Events API로 동일 ReAct 엔진을 채널 확장합니다 (`arc-reactor/README.ko.md`의 채널 구성).
- Aslan-Verse-Web은 실험형 시뮬레이션 좌표로서, 조직·페르소나·태스크 네트워크 구조를 Arc Reactor 페르소나/툴 정책 모델과 결합해 팀형 실험을 구성하는 대상군으로 정의합니다 (`Aslan-Verse-Web/docs/superpowers/specs/2026-04-05-aslan-verse-web-fe-design.md`).

3-1) 아슬란 통합 철학(흩어진 업무의 수렴)

- 기본 철학: 업무 도메인(인증, 커뮤니티, 클리핑, 보고/요약, 협업 자동화 등)이 분산되어 있어도, Arc Reactor를 중심으로 인증·도구·채널·운영 정책을 통합해 하나의 운영 체계로 묶습니다.
- 현재 근거:
  - 인증은 `aslan-iam`이 담당하고 Reactor는 공개키 기반 인증 토폴로지로 정책 수용 경로를 둡니다.
  - 작업 도구는 `POST /api/mcp/servers` 기반의 런타임 등록으로 공통 패턴에 편입됩니다.
  - 채널(웹/Slack)과 관리(운영콘솔)는 동일 정책 언어(guard/hook/tool policy)로 구동되어 조직 단위의 일관성을 유지합니다.
- 확장 로드맵: Reactor Work/Workflow를 빌더로 제공해, 팀 단위의 업무 흐름(예: 아이디어 수집→요약→승인→실행→리포팅)을 템플릿화하면 서비스별 도메인 로직만 붙여 재사용할 수 있습니다.
- 아슬란 세계관 확장 방향(계획 반영): 향후 신규 서비스가 추가될 경우, 각 서비스는 먼저 "MCP 서버 등록 방식 + Persona/Policy 설계 + Reactor 관찰 지표"의 3단계로 결합하면 기존 생태계 규칙을 유지하면서 확장할 수 있습니다.
- 의사결정 포인트: 향후 확장 시 “새 서비스마다 개별 보안 스택”보다 “Reactor 기반 정책 표준화”를 우선할지, 아니면 초기 PoC는 레이어별 게이트를 분리 적용할지 판단이 필요합니다.

4) 연동 포인트 근거 추출

- arc-reactor MCP 등록 포트/예시: curl -X POST http://localhost:18081/api/mcp/servers
- arc-reactor MCP allowlist 가드: | ARC_REACTOR_MCP_ALLOWED_SERVER_NAMES | Comma-separated MCP server allowlist (e.g. atlassian,swagger) |
- MCP 방식 운영 포인트(REST 등록, 접근 정책/Preflight): POST /api/mcp/servers, PUT /api/mcp/servers/{name}로 등록하고 갱신합니다.
- Aslan-iam 인증 키 기반 검증: JWT 토큰 발급의 유일한 권한을 가지며, 타 서비스는 공개키로 로컬 검증만 수행합니다.
- 실험 플랫폼 연결 근거: 가상 휴넷(Virtual Human Network) — AI 페르소나들이 실제 회사처럼 역할을 맡아 협업하는 플랫폼. 사용자는 AI 팀원과 함께 실무를 진행하고, 조직 구조와 업무 현황을 시각화하여 모니터링한다.

5) 주의/검증 항목(자동 보고)

- `arc-reactor`는 현재 문서 스냅샷 기준으로 동적 MCP 등록/관리 API와 스케줄러/가드 설정이 구성되어 있음이 확인됩니다.
- 프로젝트별 README 존재 유무만으로 상태를 판정합니다. 문서 기반 근거가 변경되면 다음 실행에 반영됩니다.
- 더 정확한 실시간 상태(헬스/연동 테스트)는 현재 cron 스크립트 범위 밖입니다. 필요 시 별도 운영 모니터링 Job에 연결하세요.

6) 코드/설정 정합성 근거(자동 수집)

arc-reactor 코드/설정 근거

- 저장소: /Users/jinan/ai/arc-reactor
- 근거 패턴: `@RestController|@Controller`
  - /Users/jinan/ai/arc-reactor/examples/chatbot/ChatbotExample.kt:38: * @RestController
  - /Users/jinan/ai/arc-reactor/arc-slack/src/main/kotlin/com/arc/reactor/slack/controller/SlackEventController.kt:34:@RestController
  - /Users/jinan/ai/arc-reactor/arc-slack/src/main/kotlin/com/arc/reactor/slack/controller/SlackCommandController.kt:29:@RestController
  - /Users/jinan/ai/arc-reactor/arc-admin/src/main/kotlin/com/arc/reactor/admin/controller/TenantAdminController.kt:42:@RestController
- 근거 패턴: `class .*Guard`
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/tool/idempotency/InMemoryToolIdempotencyGuard.kt:21:class InMemoryToolIdempotencyGuard(
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/guard/blockrate/GuardBlockRateHook.kt:26:class GuardBlockRateHook(
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/guard/blockrate/GuardBlockRateAnomaly.kt:8:enum class GuardAnomalyType {
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/guard/blockrate/GuardBlockRateAnomaly.kt:27:data class GuardBlockRateAnomaly(
- 근거 패턴: `class .*Hook`
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/promptlab/hook/ExperimentCaptureHook.kt:50:class ExperimentCaptureHook(
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/hook/Hook.kt:41: * class AuditHook : AfterAgentCompleteHook {
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/hook/model/HookModels.kt:16:sealed class HookResult {
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/hook/model/HookModels.kt:25: data class Reject(val reason: String) : HookResult()
- 근거 패턴: `class .*Scheduler`
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/promptlab/autoconfigure/PromptLabConfiguration.kt:185: class SchedulerConfiguration {
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/promptlab/PromptLabScheduler.kt:26:class PromptLabScheduler(
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/SchedulerController.kt:57:class SchedulerController(
  - /Users/jinan/ai/arc-reactor/arc-core/src/test/kotlin/com/arc/reactor/promptlab/PromptLabSchedulerTest.kt:26:class PromptLabSchedulerTest {
- 근거 패턴: `class .*Tool`
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/approval/ApprovalModels.kt:20:data class ToolApprovalRequest(
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/approval/ApprovalModels.kt:50:data class ToolApprovalResponse(
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/approval/ToolApprovalPolicy.kt:12: * class DestructiveToolPolicy : ToolApprovalPolicy {
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/approval/ToolApprovalPolicy.kt:23: * class AmountPolicy : ToolApprovalPolicy {
- 근거 패턴: `maxToolCalls`
  - /Users/jinan/ai/arc-reactor/helm/arc-reactor/values-production.yaml:58: maxToolCalls: "10"
  - /Users/jinan/ai/arc-reactor/helm/arc-reactor/values.yaml:162: maxToolCalls: "10"
  - /Users/jinan/ai/arc-reactor/helm/arc-reactor/templates/configmap.yaml:50: ARC_REACTOR_MAX_TOOL_CALLS: {{ .Values.config.agent.maxToolCalls | quote }}
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/intent/model/IntentModels.kt:56: * @param maxToolCalls 최대 도구 호출 수 오버라이드
- 근거 패턴: `max-tools-per-request`
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/resources/application.yml:56: max-tools-per-request: 30
  - /Users/jinan/ai/arc-reactor/arc-core/src/test/resources/application.yml:19: max-tools-per-request: 20

- 근거 패턴: `/api/mcp/servers`
  - /Users/jinan/ai/arc-reactor/arc-web/src/test/kotlin/com/arc/reactor/controller/AdminCapabilitiesControllerTest.kt:49: RequestMappingInfo.paths("/api/mcp/servers/{name}/preflight").build() to mockk()
  - /Users/jinan/ai/arc-reactor/arc-web/src/test/kotlin/com/arc/reactor/controller/AdminCapabilitiesControllerTest.kt:64: assertTrue(body.paths.contains("/api/mcp/servers/{name}/preflight")) {
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/McpSwaggerCatalogController.kt:49:@RequestMapping("/api/mcp/servers/{name}/swagger/sources")
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/McpPreflightController.kt:37:@RequestMapping("/api/mcp/servers/{name}/preflight")
- 근거 패턴: `ARC_REACTOR_MCP_ALLOWED_SERVER_NAMES`
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/resources/application.yml:208: allowed-server-names: ${ARC_REACTOR_MCP_ALLOWED_SERVER_NAMES:atlassian,swagger}
- 근거 패턴: `SSE`
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/cache/CacheMetricsRecorder.kt:78: private val missCounter: Counter = Counter.builder(METRIC_CACHE_MISSES)
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/cache/CacheMetricsRecorder.kt:112: private const val METRIC_CACHE_MISSES = "arc.cache.misses"
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/mcp/model/McpModels.kt:12: * @param transportType 전송 프로토콜 유형 (STDIO, SSE, HTTP)
  - /Users/jinan/ai/arc-reactor/arc-core/src/main/kotlin/com/arc/reactor/mcp/model/McpModels.kt:13: * @param config 전송별 설정 (예: SSE의 "url", STDIO의 "command"/"args")

aslan-iam 코드/설정 근거

- 저장소: /Users/jinan/ai/aslan-iam
- 근거 패턴: `@RestController|@Controller`
  - /Users/jinan/ai/aslan-iam/iam-api/src/main/kotlin/com/aslan/iam/api/exception/GlobalExceptionHandler.kt:18:@RestControllerAdvice
  - /Users/jinan/ai/aslan-iam/iam-admin-api/src/main/kotlin/com/aslan/iam/admin/exception/AdminExceptionHandler.kt:14:@RestControllerAdvice
  - /Users/jinan/ai/aslan-iam/iam-api/src/main/kotlin/com/aslan/iam/api/controller/TwoFactorController.kt:12:@RestController
  - /Users/jinan/ai/aslan-iam/iam-api/src/main/kotlin/com/aslan/iam/api/controller/AdminController.kt:19:@RestController
- 근거 패턴: `class .*Guard`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Hook`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Scheduler`
  - /Users/jinan/ai/aslan-iam/iam-api/src/main/kotlin/com/aslan/iam/config/InactiveAccountScheduler.kt:11:class InactiveAccountScheduler(
- 근거 패턴: `class .*Tool`
  - (해당 패턴 미발견)
- 근거 패턴: `maxToolCalls`
  - (해당 패턴 미발견)
- 근거 패턴: `max-tools-per-request`
  - (해당 패턴 미발견)
- 근거 패턴: `/api/mcp/servers`
  - (해당 패턴 미발견)
- 근거 패턴: `ARC_REACTOR_MCP_ALLOWED_SERVER_NAMES`
  - (해당 패턴 미발견)

- 근거 패턴: `SSE`
  - (해당 패턴 미발견)

swagger-mcp-server 코드/설정 근거

- 저장소: /Users/jinan/ai/swagger-mcp-server
- 근거 패턴: `@RestController|@Controller`
  - /Users/jinan/ai/swagger-mcp-server/src/main/kotlin/com/swagger/mcpserver/controller/AdminApiExceptionHandler.kt:14:@RestControllerAdvice(assignableTypes = [SpecSourceAdminController::class])
  - /Users/jinan/ai/swagger-mcp-server/src/main/kotlin/com/swagger/mcpserver/controller/AdminPreflightController.kt:16:@RestController
  - /Users/jinan/ai/swagger-mcp-server/src/main/kotlin/com/swagger/mcpserver/controller/SpecSourceAdminController.kt:25:@RestController
  - /Users/jinan/ai/swagger-mcp-server/src/main/kotlin/com/swagger/mcpserver/controller/AdminAccessPolicyController.kt:18:@RestController
- 근거 패턴: `class .*Guard`
  - /Users/jinan/ai/swagger-mcp-server/src/test/kotlin/com/swagger/mcpserver/ArchitectureGuardTest.kt:8:class ArchitectureGuardTest {
- 근거 패턴: `class .*Hook`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Scheduler`
  - /Users/jinan/ai/swagger-mcp-server/src/main/kotlin/com/swagger/mcpserver/catalog/SpecSourceScheduler.kt:16:class SpecSourceScheduler(
  - /Users/jinan/ai/swagger-mcp-server/src/main/kotlin/com/swagger/mcpserver/config/SchedulerConfig.kt:8:class SchedulerConfig {
- 근거 패턴: `class .*Tool`
  - /Users/jinan/ai/swagger-mcp-server/src/main/kotlin/com/swagger/mcpserver/tool/SpecSchemaTool.kt:9:class SpecSchemaTool(
  - /Users/jinan/ai/swagger-mcp-server/src/main/kotlin/com/swagger/mcpserver/tool/CatalogListRevisionsTool.kt:9:class CatalogListRevisionsTool(
  - /Users/jinan/ai/swagger-mcp-server/src/main/kotlin/com/swagger/mcpserver/tool/SpecLoadTool.kt:17:class SpecLoadTool(
  - /Users/jinan/ai/swagger-mcp-server/src/main/kotlin/com/swagger/mcpserver/tool/SpecRemoveTool.kt:11:class SpecRemoveTool(
- 근거 패턴: `maxToolCalls`
  - (해당 패턴 미발견)
- 근거 패턴: `max-tools-per-request`
  - (해당 패턴 미발견)
- 근거 패턴: `/api/mcp/servers`
  - (해당 패턴 미발견)
- 근거 패턴: `ARC_REACTOR_MCP_ALLOWED_SERVER_NAMES`
  - (해당 패턴 미발견)
- 근거 패턴: `SSE`
  - /Users/jinan/ai/swagger-mcp-server/build.gradle.kts:35: // Spring AI MCP Server (SSE via WebFlux)

atlassian-mcp-server 코드/설정 근거

- 저장소: /Users/jinan/ai/atlassian-mcp-server
- 근거 패턴: `@RestController|@Controller`
  - /Users/jinan/ai/atlassian-mcp-server/src/main/kotlin/com/atlassian/mcpserver/AdminPreflightController.kt:27:@RestController
  - /Users/jinan/ai/atlassian-mcp-server/src/main/kotlin/com/atlassian/mcpserver/AdminAccessPolicyController.kt:24:@RestController

- 근거 패턴: `class .*Guard`
  - /Users/jinan/ai/atlassian-mcp-server/src/main/kotlin/com/atlassian/mcpserver/tool/work/WorkPersonalInterruptGuardTool.kt:19:class WorkPersonalInterruptGuardTool(
  - /Users/jinan/ai/atlassian-mcp-server/src/test/kotlin/com/atlassian/mcpserver/tool/WorkPersonalInterruptGuardToolTest.kt:30:class WorkPersonalInterruptGuardToolTest {
- 근거 패턴: `class .*Hook`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Scheduler`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Tool`
  - /Users/jinan/ai/atlassian-mcp-server/src/main/kotlin/com/atlassian/mcpserver/tool/bitbucket/BitbucketRepoTool.kt:19:class BitbucketRepoTool(
  - /Users/jinan/ai/atlassian-mcp-server/src/main/kotlin/com/atlassian/mcpserver/tool/bitbucket/BitbucketPRTool.kt:23:class BitbucketPRTool(
  - /Users/jinan/ai/atlassian-mcp-server/src/main/kotlin/com/atlassian/mcpserver/tool/confluence/ConfluenceSearchTool.kt:20:class ConfluenceSearchTool(
  - /Users/jinan/ai/atlassian-mcp-server/src/main/kotlin/com/atlassian/mcpserver/tool/confluence/ConfluencePageTool.kt:26:class ConfluencePageTool(
- 근거 패턴: `maxToolCalls`
  - (해당 패턴 미발견)
- 근거 패턴: `max-tools-per-request`
  - (해당 패턴 미발견)
- 근거 패턴: `/api/mcp/servers`
  - (해당 패턴 미발견)
- 근거 패턴: `ARC_REACTOR_MCP_ALLOWED_SERVER_NAMES`
  - (해당 패턴 미발견)
- 근거 패턴: `SSE`
  - /Users/jinan/ai/atlassian-mcp-server/build.gradle.kts:29: // Spring AI MCP Server (SSE via Servlet/WebMVC)

clipping-mcp-server 코드/설정 근거

- 저장소: /Users/jinan/ai/clipping-mcp-server
- 근거 패턴: `@RestController|@Controller`
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/config/SpaController.kt:10:@RestController
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/admin/CategoryRuleAdminController.kt:21:@RestController
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/admin/KeywordTrendAdminController.kt:13:@RestController
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/admin/UserCompetitorViewController.kt:21:@RestController
- 근거 패턴: `class .*Guard`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Hook`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Scheduler`
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/service/SlackTokenValidationScheduler.kt:19:class SlackTokenValidationScheduler(
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/service/EmptyCategoryScheduler.kt:21:class EmptyCategoryScheduler(
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/service/SystemStatusService.kt:122: data class SchedulerInfo(
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/config/SchedulerConfig.kt:19:class SchedulerConfig {

- 근거 패턴: `class .*Tool`
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/tool/ClipCollectAsyncTool.kt:9:class ClipCollectAsyncTool(private val asyncClipJobService: AsyncClipJobService) {
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/tool/ClipExportTool.kt:9:class ClipExportTool(private val clippingService: ClippingService) {
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/tool/CategoryListTool.kt:9:class CategoryListTool(private val categoryService: CategoryService) {
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/tool/ClipDigestTool.kt:9:class ClipDigestTool(private val clippingService: ClippingService) {
- 근거 패턴: `maxToolCalls`
  - (해당 패턴 미발견)
- 근거 패턴: `max-tools-per-request`
  - (해당 패턴 미발견)
- 근거 패턴: `/api/mcp/servers`
  - (해당 패턴 미발견)
- 근거 패턴: `ARC_REACTOR_MCP_ALLOWED_SERVER_NAMES`
  - (해당 패턴 미발견)
- 근거 패턴: `SSE`
  - /Users/jinan/ai/clipping-mcp-server/build.gradle.kts:38: // Spring AI MCP Server (SSE via WebFlux)
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/service/UserAccountApprovalService.kt:272: val code = if (e.message?.contains("상태") == true) "ALREADY_PROCESSED" else "INVALID_ROLE"
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/service/UserAccountApprovalService.kt:326: val code = if (e.message?.contains("상태") == true) "ALREADY_PROCESSED" else "INVALID_ROLE"
  - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/admin/AdminUserRequestController.kt:137: is com.clipping.mcpserver.error.InvalidInputException -> "ALREADY_PROCESSED"

arc-reactor-admin 코드/설정 근거

- 저장소: /Users/jinan/ai/arc-reactor-admin
- 근거 패턴: `@RestController|@Controller`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Guard`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Hook`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Scheduler`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Tool`
  - (해당 패턴 미발견)
- 근거 패턴: `maxToolCalls`
  - (해당 패턴 미발견)
- 근거 패턴: `max-tools-per-request`
  - (해당 패턴 미발견)
- 근거 패턴: `/api/mcp/servers`
  - (해당 패턴 미발견)

- 근거 패턴: `ARC_REACTOR_MCP_ALLOWED_SERVER_NAMES`
  - (해당 패턴 미발견)
- 근거 패턴: `SSE`
  - /Users/jinan/ai/arc-reactor-admin/pnpm-lock.yaml:1997: resolution: {integrity: sha512-1to4zXBxmXHV3IiSSEInrreIlu02vUOvrhxJJH5vcxYTBDAx51cqZiKdyTxlecdKNSjj8EcxGBxNf6Vg+945gw==}

arc-reactor/arc-web 코드/설정 근거

- 저장소: /Users/jinan/ai/arc-reactor/arc-web
- 근거 패턴: `@RestController|@Controller`
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/SchedulerController.kt:54:@RestController
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/OutputGuardRuleController.kt:53:@RestController
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/RagIngestionCandidateController.kt:43:@RestController
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/McpSwaggerCatalogController.kt:48:@RestController
- 근거 패턴: `class .*Guard`
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/OutputGuardRuleController.kt:61:class OutputGuardRuleController(
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/OutputGuardRuleController.kt:298:data class CreateOutputGuardRuleRequest(
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/OutputGuardRuleController.kt:320:data class UpdateOutputGuardRuleRequest(
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/OutputGuardRuleController.kt:332:data class OutputGuardSimulationRequest(
- 근거 패턴: `class .*Hook`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Scheduler`
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/SchedulerController.kt:57:class SchedulerController(
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/OpsDashboardController.kt:396:data class SchedulerOpsSummary(
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/OpsDashboardController.kt:405:data class RecentSchedulerExecutionSummary(
  - /Users/jinan/ai/arc-reactor/arc-web/src/test/kotlin/com/arc/reactor/controller/SchedulerControllerConditionalTest.kt:19:class SchedulerControllerConditionalTest {
- 근거 패턴: `class .*Tool`
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/ToolPolicyController.kt:41:class ToolPolicyController(
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/ToolPolicyController.kt:122:data class ToolPolicyStateResponse(
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/ToolPolicyController.kt:129:data class ToolPolicyResponse(
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/ToolPolicyController.kt:140:data class UpdateToolPolicyRequest(
- 근거 패턴: `maxToolCalls`
  - /Users/jinan/ai/arc-reactor/arc-web/src/test/kotlin/com/arc/reactor/integration/IntentControllerJdbcIntegrationTest.kt:106: "maxToolCalls" to 5,
- 근거 패턴: `max-tools-per-request`
  - (해당 패턴 미발견)
- 근거 패턴: `/api/mcp/servers`
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/McpSwaggerCatalogController.kt:49:@RequestMapping("/api/mcp/servers/{name}/swagger/sources")
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/McpPreflightController.kt:37:@RequestMapping("/api/mcp/servers/{name}/preflight")
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/McpAccessPolicyController.kt:42:@RequestMapping("/api/mcp/servers/{name}/access-policy")
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/McpServerController.kt:44: * - GET /api/mcp/servers : 전체 서버 목록 및 상태 조회

- 근거 패턴: `ARC_REACTOR_MCP_ALLOWED_SERVER_NAMES`
  - (해당 패턴 미발견)
- 근거 패턴: `SSE`
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/ChatController.kt:54: * - POST /api/chat/stream : 스트리밍 응답 (SSE, 실시간 토큰 단위 전송)
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/ChatController.kt:56: * ## SSE 이벤트 타입 (스트리밍)
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/ChatController.kt:117: * 스트리밍 채팅 -- 타입이 지정된 SSE 이벤트를 통한 실시간 응답.
  - /Users/jinan/ai/arc-reactor/arc-web/src/main/kotlin/com/arc/reactor/controller/ChatController.kt:119: * SSE 이벤트 타입:

arc-reactor/arc-slack 코드/설정 근거

- 저장소: /Users/jinan/ai/arc-reactor/arc-slack
- 근거 패턴: `@RestController|@Controller`
  - /Users/jinan/ai/arc-reactor/arc-slack/src/main/kotlin/com/arc/reactor/slack/controller/SlackEventController.kt:34:@RestController
  - /Users/jinan/ai/arc-reactor/arc-slack/src/main/kotlin/com/arc/reactor/slack/controller/SlackCommandController.kt:29:@RestController
- 근거 패턴: `class .*Guard`
  - /Users/jinan/ai/arc-reactor/arc-slack/src/test/kotlin/com/arc/reactor/slack/session/SlackThreadTrackerTest.kt:50: inner class BlankInputGuard {
- 근거 패턴: `class .*Hook`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Scheduler`
  - /Users/jinan/ai/arc-reactor/arc-slack/src/main/kotlin/com/arc/reactor/slack/handler/SlackReminderScheduler.kt:27:class SlackReminderScheduler(
  - /Users/jinan/ai/arc-reactor/arc-slack/src/test/kotlin/com/arc/reactor/slack/handler/SlackReminderSchedulerTest.kt:19:class SlackReminderSchedulerTest {
- 근거 패턴: `class .*Tool`
  - /Users/jinan/ai/arc-reactor/arc-slack/src/main/kotlin/com/arc/reactor/slack/tools/tool/CreateCanvasTool.kt:9:open class CreateCanvasTool(
  - /Users/jinan/ai/arc-reactor/arc-slack/src/main/kotlin/com/arc/reactor/slack/tools/observability/ToolObservabilityAspect.kt:27:class ToolObservabilityAspect(
  - /Users/jinan/ai/arc-reactor/arc-slack/src/main/kotlin/com/arc/reactor/slack/tools/observability/ToolObservabilityAspect.kt:125: private data class ToolOutcome(
  - /Users/jinan/ai/arc-reactor/arc-slack/src/main/kotlin/com/arc/reactor/slack/tools/tool/ReadThreadRepliesTool.kt:9:open class ReadThreadRepliesTool(private val readThreadRepliesUseCase: ReadThreadRepliesUseCase) : LocalTool {
- 근거 패턴: `maxToolCalls`
  - (해당 패턴 미발견)
- 근거 패턴: `max-tools-per-request`
  - (해당 패턴 미발견)
- 근거 패턴: `/api/mcp/servers`
  - (해당 패턴 미발견)
- 근거 패턴: `ARC_REACTOR_MCP_ALLOWED_SERVER_NAMES`
  - (해당 패턴 미발견)

- 근거 패턴: `SSE`
  - /Users/jinan/ai/arc-reactor/arc-slack/src/test/kotlin/com/arc/reactor/slack/SlackCrossToolAndProactiveE2ETest.kt:53: McpServer(name = name, transportType = McpTransportType.SSE, config = emptyMap())
  - /Users/jinan/ai/arc-reactor/arc-slack/src/test/kotlin/com/arc/reactor/slack/DefaultSlackEventHandlerProactiveTest.kt:63: McpServer(name = name, transportType = McpTransportType.SSE, config = emptyMap())
  - /Users/jinan/ai/arc-reactor/arc-slack/src/test/kotlin/com/arc/reactor/slack/handler/SlackHandlerSupportTest.kt:27: /** 테스트용 SSE 기반 McpServer 팩토리. */
  - /Users/jinan/ai/arc-reactor/arc-slack/src/test/kotlin/com/arc/reactor/slack/handler/SlackHandlerSupportTest.kt:30: transportType = McpTransportType.SSE,

arc-reactor-web 코드/설정 근거

- 저장소: /Users/jinan/ai/arc-reactor-web
- 근거 패턴: `@RestController|@Controller`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Guard`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Hook`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Scheduler`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Tool`
  - (해당 패턴 미발견)
- 근거 패턴: `maxToolCalls`
  - (해당 패턴 미발견)
- 근거 패턴: `max-tools-per-request`
  - (해당 패턴 미발견)
- 근거 패턴: `/api/mcp/servers`
  - (해당 패턴 미발견)
- 근거 패턴: `ARC_REACTOR_MCP_ALLOWED_SERVER_NAMES`
  - (해당 패턴 미발견)
- 근거 패턴: `SSE`
  - /Users/jinan/ai/arc-reactor-web/pnpm-lock.yaml:2044: resolution: {integrity: sha512-1to4zXBxmXHV3IiSSEInrrellu02vUOvrhxJJH5vcxYTBDAx51cqZiKdyTxlecdKNSjj8EcxGBxNf6Vg+945gw==}

Aslan-Verse-Web 코드/설정 근거

- 저장소: /Users/jinan/ai/Aslan-Verse-Web
- 근거 패턴: `@RestController|@Controller`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Guard`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Hook`
  - (해당 패턴 미발견)

- 근거 패턴: `class .*Scheduler`
  - (해당 패턴 미발견)
- 근거 패턴: `class .*Tool`
  - (해당 패턴 미발견)
- 근거 패턴: `maxToolCalls`
  - (해당 패턴 미발견)
- 근거 패턴: `max-tools-per-request`
  - (해당 패턴 미발견)
- 근거 패턴: `/api/mcp/servers`
  - (해당 패턴 미발견)
- 근거 패턴: `ARC_REACTOR_MCP_ALLOWED_SERVER_NAMES`
  - (해당 패턴 미발견)
- 근거 패턴: `SSE`
  - (해당 패턴 미발견)

7) API 구현 근거(Controller/Route)

arc-reactor API 엔드포인트 근거

- 저장소: /Users/jinan/ai/arc-reactor
  - examples/chatbot/ChatbotExample.kt
    - 41: * @PostMapping("/chat") -> 42: * suspend fun chat(@RequestBody request: ChatRequest): ChatResponse {
      - 42: * suspend fun chat(@RequestBody request: ChatRequest): ChatResponse {
  - arc-admin/src/main/kotlin/com/arc/reactor/admin/controller/TenantAdminController.kt
    - 47:@RequestMapping("/api/admin/tenant")
      - 59: @Operation(summary = "Get tenant overview dashboard")
    - 65: @GetMapping("/overview") -> 66: fun overview(exchange: ServerWebExchange): ResponseEntity {
      - 66: fun overview(exchange: ServerWebExchange): ResponseEntity {
    - 80: @GetMapping("/usage") -> 81: fun usage(
      - 81: fun usage(
    - 98: @GetMapping("/quality") -> 99: fun quality(
      - 99: fun quality(
    - 116: @GetMapping("/tools") -> 117: fun tools(
      - 117: fun tools(
    - 134: @GetMapping("/cost") -> 135: fun cost(
      - 135: fun cost(
    - 153: @GetMapping("/slo") -> 154: fun slo(exchange: ServerWebExchange): ResponseEntity {
      - 154: fun slo(exchange: ServerWebExchange): ResponseEntity {
    - 174: @GetMapping("/alerts") -> 175: fun alerts(exchange: ServerWebExchange): ResponseEntity {
      - 175: fun alerts(exchange: ServerWebExchange): ResponseEntity {
    - 188: @GetMapping("/quota") -> 189: fun quota(exchange: ServerWebExchange): ResponseEntity {
      - 189: fun quota(exchange: ServerWebExchange): ResponseEntity {
    - 213: @GetMapping("/export/executions") -> 214: fun exportExecutions(
      - 214: fun exportExecutions(
    - 242: @GetMapping("/export/tools") -> 243: fun exportTools(
      - 243: fun exportTools(

- arc-admin/src/main/kotlin/com/arc/reactor/admin/controller/PlatformAdminController.kt
  - 63:@RequestMapping("/api/admin/platform")
  - 89: @GetMapping("/health") -> 90: fun health(exchange: ServerWebExchange): ResponseEntity {
    - 90: fun health(exchange: ServerWebExchange): ResponseEntity {
  - 115: @GetMapping("/users/by-email") -> 116: fun getUserByEmail(
    - 116: fun getUserByEmail(
  - 139: @PostMapping("/users/{id}/role") -> 140: fun updateUserRole(
    - 140: fun updateUserRole(
  - 175: @GetMapping("/tenants") -> 176: fun listTenants(exchange: ServerWebExchange): ResponseEntity {
    - 176: fun listTenants(exchange: ServerWebExchange): ResponseEntity {
  - 188: @GetMapping("/tenants/{id}") -> 189: fun getTenant(
    - 189: fun getTenant(
  - 206: @PostMapping("/tenants") -> 207: fun createTenant(
    - 207: fun createTenant(
  - 242: @PostMapping("/tenants/{id}/suspend") -> 243: fun suspendTenant(
    - 243: fun suspendTenant(
  - 271: @PostMapping("/tenants/{id}/activate") -> 272: fun activateTenant(
    - 272: fun activateTenant(
  - 299: @GetMapping("/tenants/analytics") -> 300: fun tenantAnalytics(exchange: ServerWebExchange): ResponseEntity {
    - 300: fun tenantAnalytics(exchange: ServerWebExchange): ResponseEntity {
  - 333: @GetMapping("/pricing") -> 334: fun listPricing(exchange: ServerWebExchange): ResponseEntity {
    - 334: fun listPricing(exchange: ServerWebExchange): ResponseEntity {
  - 345: @PostMapping("/pricing") -> 346: fun upsertPricing(
    - 346: fun upsertPricing(
  - 370: @PostMapping("/cache/invalidate") -> 371: fun invalidateResponseCache(exchange: ServerWebExchange): ResponseEntity {
    - 371: fun invalidateResponseCache(exchange: ServerWebExchange): ResponseEntity {
  - 414: @GetMapping("/alerts/rules") -> 415: fun listAlertRules(exchange: ServerWebExchange): ResponseEntity {
    - 415: fun listAlertRules(exchange: ServerWebExchange): ResponseEntity {
  - 426: @PostMapping("/alerts/rules") -> 427: fun saveAlertRule(
    - 427: fun saveAlertRule(
  - 451: @DeleteMapping("/alerts/rules/{id}") -> 452: fun deleteAlertRule(
    - 452: fun deleteAlertRule(
  - 478: @GetMapping("/alerts") -> 479: fun activeAlerts(exchange: ServerWebExchange): ResponseEntity {
    - 479: fun activeAlerts(exchange: ServerWebExchange): ResponseEntity {
  - 490: @PostMapping("/alerts/{id}/resolve") -> 491: fun resolveAlert(
    - 491: fun resolveAlert(
  - 514: @PostMapping("/alerts/evaluate") -> 515: fun evaluateAlerts(exchange: ServerWebExchange): ResponseEntity {
    - 515: fun evaluateAlerts(exchange: ServerWebExchange): ResponseEntity {
  - 536: @GetMapping("/cache/stats") -> 537: fun cacheStats(exchange: ServerWebExchange): ResponseEntity {
    - 537: fun cacheStats(exchange: ServerWebExchange): ResponseEntity {
  - 572: @GetMapping("/vectorstore/stats") -> 573: fun vectorStoreStats(exchange: ServerWebExchange): ResponseEntity {
    - 573: fun vectorStoreStats(exchange: ServerWebExchange): ResponseEntity {
- arc-admin/src/main/kotlin/com/arc/reactor/admin/controller/MetricIngestionController.kt
  - 39:@RequestMapping("/api/admin/metrics/ingest")
    - 50: @Operation(summary = "Ingest MCP health event")
    - 57: fun ingestMcpHealth(
  - 56: @PostMapping("/mcp-health") -> 57: fun ingestMcpHealth(
    - 57: fun ingestMcpHealth(
  - 82: @PostMapping("/tool-call") -> 83: fun ingestToolCall(
    - 83: fun ingestToolCall(
  - 111: @PostMapping("/eval-result") -> 112: fun ingestEvalResult(
    - 112: fun ingestEvalResult(
  - 127: @PostMapping("/eval-results") -> 128: fun ingestEvalResults(
    - 128: fun ingestEvalResults(
  - 189: @PostMapping("/batch") -> 190: fun ingestBatch(
    - 190: fun ingestBatch(

- arc-admin/src/main/kotlin/com/arc/reactor/admin/controller/AdminSessionController.kt
  - 44:@RequestMapping("/api/admin")
    - 52: @Operation(summary = "대화 Overview 통계 조회")
    - 58: fun getOverview(
  - 57: @GetMapping("/sessions/overview") -> 58: fun getOverview(
    - 58: fun getOverview(
  - 76: @GetMapping("/sessions") -> 77: fun listSessions(
    - 77: fun listSessions(
  - 111: @GetMapping("/sessions/{sessionId}") -> 112: fun getSessionDetail(
    - 112: fun getSessionDetail(
  - 131: @DeleteMapping("/sessions/{sessionId}") -> 132: fun deleteSession(
    - 132: fun deleteSession(
  - 161: @GetMapping("/sessions/{sessionId}/export") -> 162: fun exportSession(
    - 162: fun exportSession(
  - 218: @GetMapping("/users") -> 219: fun listUsers(
    - 219: fun listUsers(
  - 242: @GetMapping("/users/{userId}/sessions") -> 243: fun listUserSessions(
    - 243: fun listUserSessions(
  - 277: @PostMapping("/sessions/{sessionId}/tags") -> 278: fun addTag(
    - 278: fun addTag(
  - 300: @DeleteMapping("/sessions/{sessionId}/tags/{tagId}") -> 301: fun removeTag(
    - 301: fun removeTag(
- arc-slack/src/main/kotlin/com/arc/reactor/slack/controller/SlackEventController.kt
  - 35:@RequestMapping("/api/slack")
    - 39:@Tag(name = "Slack", description = "Slack 이벤트 웹훅 엔드포인트")
    - 45: @Operation(summary = "Slack 이벤트 콜백 수신 (URL 검증 포함)")
    - 50: suspend fun handleEvent(
  - 44: @PostMapping("/events") -> 50: suspend fun handleEvent(
    - 45: @Operation(summary = "Slack 이벤트 콜백 수신 (URL 검증 포함)")
    - 50: suspend fun handleEvent(
- arc-slack/src/main/kotlin/com/arc/reactor/slack/controller/SlackCommandController.kt
  - 30:@RequestMapping("/api/slack")
    - 34:@Tag(name = "Slack", description = "Slack 슬래시 명령 처리 엔드포인트")
  - 49: @PostMapping("/commands", consumes = [MediaType.APPLICATION_FORM_URLENCODED_VALUE]) -> 55: fun handleSlashCommandEndpoint(
    - 50: @Operation(summary = "Slack 슬래시 명령 수신 및 처리")
    - 55: fun handleSlashCommandEndpoint(
- arc-web/src/main/kotlin/com/arc/reactor/controller/SchedulerController.kt
  - 55:@RequestMapping("/api/scheduler/jobs")
    - 62: @Operation(summary = "전체 예약 작업 목록 조회 (태그 필터 선택)")
    - 68: fun listJobs(
  - 67: @GetMapping -> 68: fun listJobs(
    - 68: fun listJobs(
  - 88: @PostMapping -> 89: fun createJob(
    - 89: fun createJob(
  - 111: @GetMapping("/{id}") -> 112: fun getJob(@PathVariable id: String, exchange: ServerWebExchange): ResponseEntity {
    - 112: fun getJob(@PathVariable id: String, exchange: ServerWebExchange): ResponseEntity {
  - 126: @PutMapping("/{id}") -> 127: fun updateJob(
    - 127: fun updateJob(
  - 150: @DeleteMapping("/{id}") -> 151: fun deleteJob(@PathVariable id: String, exchange: ServerWebExchange): ResponseEntity {
    - 151: fun deleteJob(@PathVariable id: String, exchange: ServerWebExchange): ResponseEntity {
  - 165: @PostMapping("/{id}/trigger") -> 166: fun triggerJob(@PathVariable id: String, exchange: ServerWebExchange): Mono<ResponseEntity> {
    - 166: fun triggerJob(@PathVariable id: String, exchange: ServerWebExchange): Mono<ResponseEntity> {
  - 182: @PostMapping("/{id}/dry-run") -> 183: fun dryRunJob(@PathVariable id: String, exchange: ServerWebExchange): Mono<ResponseEntity> {
    - 183: fun dryRunJob(@PathVariable id: String, exchange: ServerWebExchange): Mono<ResponseEntity> {
  - 199: @GetMapping("/{id}/executions") -> 200: fun getExecutions(
    - 200: fun getExecutions(

- arc-web/src/main/kotlin/com/arc/reactor/controller/OutputGuardRuleController.kt
  - 54:@RequestMapping("/api/output-guard/rules")
  - 90: @GetMapping -> 91: fun listRules(exchange: ServerWebExchange): ResponseEntity {
    - 91: fun listRules(exchange: ServerWebExchange): ResponseEntity {
  - 103: @GetMapping("/audits") -> 104: fun listAudits(
    - 104: fun listAudits(
  - 122: @PostMapping -> 123: fun createRule(
    - 123: fun createRule(
  - 165: @PutMapping("/{id}") -> 166: fun updateRule(
    - 166: fun updateRule(
  - 210: @DeleteMapping("/{id}") -> 211: fun deleteRule(
    - 211: fun deleteRule(
  - 236: @PostMapping("/simulate") -> 237: fun simulate(
    - 237: fun simulate(
- arc-web/src/main/kotlin/com/arc/reactor/controller/PersonaController.kt
  - 43:@RequestMapping("/api/personas") -> 55: fun listPersonas(
    - 49: @Operation(summary = "전체 페르소나 목록 조회 (관리자)")
    - 55: fun listPersonas(
  - 54: @GetMapping -> 55: fun listPersonas(
    - 55: fun listPersonas(
  - 71: @GetMapping("/{personaId}") -> 72: fun getPersona(@PathVariable personaId: String): ResponseEntity {
    - 72: fun getPersona(@PathVariable personaId: String): ResponseEntity {
  - 85: @PostMapping -> 86: fun createPersona(
    - 86: fun createPersona(
  - 115: @PutMapping("/{personaId}") -> 116: fun updatePersona(
    - 116: fun updatePersona(
  - 143: @DeleteMapping("/{personaId}") -> 144: fun deletePersona(
    - 144: fun deletePersona(
- arc-web/src/main/kotlin/com/arc/reactor/controller/RagIngestionCandidateController.kt
  - 44:@RequestMapping("/api/rag-ingestion/candidates")
    - 57: @Operation(summary = "RAG Ingestion 후보 목록 조회 (관리자)")
  - 62: @GetMapping -> 63: fun list(
    - 63: fun list(
  - 83: @PostMapping("/{id}/approve") -> 84: fun approve(
    - 84: fun approve(
  - 137: @PostMapping("/{id}/reject") -> 138: fun reject(
    - 138: fun reject(
- arc-web/src/main/kotlin/com/arc/reactor/controller/McpSwaggerCatalogController.kt
  - 49:@RequestMapping("/api/mcp/servers/{name}/swagger/sources")
    - 59: @Operation(summary = "MCP admin API를 통해 Swagger 스펙 소스 목록 조회")
    - 66: suspend fun listSources(
  - 65: @GetMapping -> 66: suspend fun listSources(
    - 66: suspend fun listSources(
  - 87: @GetMapping("/{sourceName}") -> 88: suspend fun getSource(
    - 88: suspend fun getSource(
  - 106: @PostMapping -> 107: suspend fun createSource(
    - 107: suspend fun createSource(
  - 124: @PutMapping("/{sourceName}") -> 125: suspend fun updateSource(
    - 125: suspend fun updateSource(
  - 144: @PostMapping("/{sourceName}/sync") -> 145: suspend fun syncSource(
    - 145: suspend fun syncSource(
  - 163: @GetMapping("/{sourceName}/revisions") -> 164: suspend fun listRevisions(
    - 164: suspend fun listRevisions(
  - 184: @GetMapping("/{sourceName}/diff") -> 185: suspend fun getDiff(
    - 185: suspend fun getDiff(
  - 209: @PostMapping("/{sourceName}/publish") -> 210: suspend fun publishRevision(
    - 210: suspend fun publishRevision(

aslan-iam API 엔드포인트 근거

- 저장소: /Users/jinan/ai/aslan-iam
  - iam-api/src/main/kotlin/com/aslan/iam/api/exception/GlobalExceptionHandler.kt
    - (매핑 애노테이션 미발견)
  - iam-api/src/main/kotlin/com/aslan/iam/api/controller/TwoFactorController.kt
    - 13:@RequestMapping("/api/2fa") -> 19: fun setup(@AuthenticationPrincipal userId: String): ResponseEntity {
      - 19: fun setup(@AuthenticationPrincipal userId: String): ResponseEntity {
    - 18: @PostMapping("/setup") -> 19: fun setup(@AuthenticationPrincipal userId: String): ResponseEntity {
      - 19: fun setup(@AuthenticationPrincipal userId: String): ResponseEntity {
    - 26: @PostMapping("/enable") -> 27: fun enable(
      - 27: fun enable(
    - 35: @PostMapping("/disable") -> 36: fun disable(
      - 36: fun disable(
  - iam-admin-api/src/main/kotlin/com/aslan/iam/admin/exception/AdminExceptionHandler.kt
    - (매핑 애노테이션 미발견)
  - iam-api/src/main/kotlin/com/aslan/iam/api/controller/AdminController.kt
    - 20:@RequestMapping("/api/admin") -> 30: fun assignRole(
      - 30: fun assignRole(
    - 29: @PostMapping("/users/{userId}/roles") -> 30: fun assignRole(
      - 30: fun assignRole(
    - 40: @DeleteMapping("/users/{userId}/roles") -> 41: fun revokeRole(
      - 41: fun revokeRole(
    - 51: @DeleteMapping("/users/{userId}/sessions") -> 52: fun forceLogout(
      - 52: fun forceLogout(
  - iam-api/src/main/kotlin/com/aslan/iam/api/controller/AuthController.kt
    - 15:@RequestMapping("/api/auth") -> 26: fun register(@Valid @RequestBody request: RegisterRequest): ResponseEntity {
      - 26: fun register(@Valid @RequestBody request: RegisterRequest): ResponseEntity {
    - 25: @PostMapping("/register") -> 26: fun register(@Valid @RequestBody request: RegisterRequest): ResponseEntity {
      - 26: fun register(@Valid @RequestBody request: RegisterRequest): ResponseEntity {
    - 35: @PostMapping("/login") -> 36: fun login(@Valid @RequestBody request: LoginRequest): ResponseEntity {
      - 36: fun login(@Valid @RequestBody request: LoginRequest): ResponseEntity {
    - 57: @PostMapping("/refresh") -> 58: fun refresh(@Valid @RequestBody request: RefreshRequest): ResponseEntity {
      - 58: fun refresh(@Valid @RequestBody request: RefreshRequest): ResponseEntity {
    - 69: @PostMapping("/logout") -> 70: fun logout(@AuthenticationPrincipal userId: String): ResponseEntity {
      - 70: fun logout(@AuthenticationPrincipal userId: String): ResponseEntity {
    - 75: @PostMapping("/change-password") -> 76: fun changePassword(
      - 76: fun changePassword(
    - 90: @GetMapping("/public-key") -> 91: fun getPublicKey(): ResponseEntity {
      - 91: fun getPublicKey(): ResponseEntity {

- iam-admin-api/src/main/kotlin/com/aslan/iam/admin/controller/AdminController.kt
  - 22:@RequestMapping("/api/admin")
  - 43: @GetMapping("/users") -> 44: fun getUsers(
    - 44: fun getUsers(
  - 67: @GetMapping("/users/{userId}") -> 68: fun getUserDetail(
    - 68: fun getUserDetail(
  - 85: @PatchMapping("/users/{userId}/status") -> 86: fun updateUserStatus(
    - 86: fun updateUserStatus(
  - 97: @DeleteMapping("/users/{userId}/2fa") -> 98: fun forceDisable2FA(
    - 98: fun forceDisable2FA(
  - 107: @PostMapping("/users/{userId}/roles") -> 108: fun assignRole(
    - 108: fun assignRole(
  - 119: @DeleteMapping("/users/{userId}/roles") -> 120: fun revokeRole(
    - 120: fun revokeRole(
  - 131: @DeleteMapping("/users/{userId}/sessions") -> 132: fun forceLogout(
    - 132: fun forceLogout(
  - 143: @PatchMapping("/users/bulk/status") -> 144: fun bulkUpdateStatus(
    - 144: fun bulkUpdateStatus(
  - 152: @DeleteMapping("/users/bulk/sessions") -> 153: fun bulkForceLogout(
    - 153: fun bulkForceLogout(
  - 161: @PostMapping("/users/bulk/roles") -> 162: fun bulkAssignRole(
    - 162: fun bulkAssignRole(
  - 172: @GetMapping("/roles") -> 173: fun getRoles(): ResponseEntity<List> {
    - 173: fun getRoles(): ResponseEntity<List> {
  - 185: @PostMapping("/roles") -> 186: fun createRole(
    - 186: fun createRole(
  - 202: @PutMapping("/roles/{roleId}") -> 203: fun updateRole(
    - 203: fun updateRole(
  - 213: @DeleteMapping("/roles/{roleId}") -> 214: fun deleteRole(
    - 214: fun deleteRole(
  - 223: @GetMapping("/permissions") -> 224: fun getPermissions(): ResponseEntity<List> {
    - 224: fun getPermissions(): ResponseEntity<List> {
  - 233: @GetMapping("/audit-logs") -> 234: fun getAuditLogs(
    - 234: fun getAuditLogs(
  - 275: @GetMapping("/dashboard/stats") -> 276: fun getDashboardStats(): ResponseEntity =
    - 276: fun getDashboardStats(): ResponseEntity =
  - 279: @GetMapping("/dashboard/login-trend") -> 280: fun getLoginTrend(@RequestParam(defaultValue = "7") days: Int): ResponseEntity<List> =
    - 280: fun getLoginTrend(@RequestParam(defaultValue = "7") days: Int): ResponseEntity<List> =
  - 285: @GetMapping("/sessions") -> 286: fun getSessions(
    - 286: fun getSessions(
  - 314: @DeleteMapping("/sessions") -> 315: fun revokeAllSessions(@AuthenticationPrincipal adminId: String): ResponseEntity {
    - 315: fun revokeAllSessions(@AuthenticationPrincipal adminId: String): ResponseEntity {
  - 323: @GetMapping("/users/export") -> 324: fun exportUsers(
    - 324: fun exportUsers(
  - 361: @GetMapping("/settings/system") -> 362: fun getSystemSettings(): ResponseEntity {
    - 362: fun getSystemSettings(): ResponseEntity {
  - 391: @GetMapping("/security-policies") -> 392: fun getSecurityPolicies(): ResponseEntity =
    - 392: fun getSecurityPolicies(): ResponseEntity =

swagger-mcp-server API 엔드포인트 근거

- 저장소: /Users/jinan/ai/swagger-mcp-server
  - src/main/kotlin/com/swagger/mcpserver/controller/AdminApiExceptionHandler.kt
    - (매핑 애노테이션 미발견)
  - src/main/kotlin/com/swagger/mcpserver/controller/AdminPreflightController.kt
    - 17:@RequestMapping("/admin/preflight") -> 27: fun preflight(): ResponseEntity {
      - 25: @Operation(summary = "Run production-readiness preflight checks for swagger MCP")
      - 27: fun preflight(): ResponseEntity {
    - 26: @GetMapping -> 27: fun preflight(): ResponseEntity {
      - 27: fun preflight(): ResponseEntity {
  - src/main/kotlin/com/swagger/mcpserver/controller/SpecSourceAdminController.kt
    - 26:@RequestMapping("/admin/spec-sources") -> 37: fun listSources(): Mono<List> = blocking {
      - 37: fun listSources(): Mono<List> = blocking {
    - 36: @GetMapping -> 37: fun listSources(): Mono<List> = blocking {
      - 37: fun listSources(): Mono<List> = blocking {
    - 41: @GetMapping("/{name}") -> 42: fun getSource(@PathVariable name: String): Mono = blocking {
      - 42: fun getSource(@PathVariable name: String): Mono = blocking {
    - 46: @PostMapping -> 47: fun createSource(@Valid @RequestBody request: CreateSpecSourceRequest): Mono<ResponseEntity> = blocking {
      - 47: fun createSource(@Valid @RequestBody request: CreateSpecSourceRequest): Mono<ResponseEntity> = blocking {
    - 63: @PutMapping("/{name}") -> 64: fun updateSource(
      - 64: fun updateSource(
    - 83: @PostMapping("/{name}/sync") -> 84: fun syncSource(@PathVariable name: String): Mono = blocking {
      - 84: fun syncSource(@PathVariable name: String): Mono = blocking {
    - 88: @GetMapping("/{name}/revisions") -> 89: fun listRevisions(
      - 89: fun listRevisions(
    - 96: @GetMapping("/{name}/diff") -> 97: fun getDiff(
      - 97: fun getDiff(
    - 105: @PostMapping("/{name}/publish") -> 106: fun publishRevision(
      - 106: fun publishRevision(
  - src/main/kotlin/com/swagger/mcpserver/controller/AdminAccessPolicyController.kt
    - 19:@RequestMapping("/admin/access-policy") -> 26: fun get(): ResponseEntity = ResponseEntity.ok(toResponse(policyStore.get()))
      - 24: @Operation(summary = "Get current swagger spec access policy")
      - 26: fun get(): ResponseEntity = ResponseEntity.ok(toResponse(policyStore.get()))
    - 25: @GetMapping -> 26: fun get(): ResponseEntity = ResponseEntity.ok(toResponse(policyStore.get()))
      - 26: fun get(): ResponseEntity = ResponseEntity.ok(toResponse(policyStore.get()))
    - 29: @PutMapping -> 30: fun put(@Valid @RequestBody request: UpdateSpecAccessPolicyRequest): ResponseEntity {
      - 30: fun put(@Valid @RequestBody request: UpdateSpecAccessPolicyRequest): ResponseEntity {
    - 49: @DeleteMapping -> 50: fun clear(): ResponseEntity {
      - 50: fun clear(): ResponseEntity {

atlassian-mcp-server API 엔드포인트 근거

- 저장소: /Users/jinan/ai/atlassian-mcp-server
  - src/main/kotlin/com/atlassian/mcpserver/AdminPreflightController.kt
    - 28:@RequestMapping("/admin/preflight")
      - 43: @Operation(
    - 47: @GetMapping -> 48: fun run(): ResponseEntity {
      - 48: fun run(): ResponseEntity {
  - src/main/kotlin/com/atlassian/mcpserver/AdminAccessPolicyController.kt
    - 25:@RequestMapping("/admin/access-policy") -> 36: fun get(): ResponseEntity {
      - 34: @Operation(summary = "Get current access policy (allowlists)")
      - 36: fun get(): ResponseEntity {
    - 35: @GetMapping -> 36: fun get(): ResponseEntity {
      - 36: fun get(): ResponseEntity {
    - 59: @PutMapping -> 60: fun put(
      - 60: fun put(
    - 88: @DeleteMapping -> 89: fun clear(httpRequest: HttpServletRequest): ResponseEntity {
      - 89: fun clear(httpRequest: HttpServletRequest): ResponseEntity {
    - 116: @PostMapping("/emergency-deny-all") -> 117: fun emergencyDenyAll(httpRequest: HttpServletRequest): ResponseEntity {
      - 117: fun emergencyDenyAll(httpRequest: HttpServletRequest): ResponseEntity {
    - 147: @GetMapping("/audits") -> 148: fun listAudits(
      - 148: fun listAudits(

clipping-mcp-server API 엔드포인트 근거

- 저장소: /Users/jinan/ai/clipping-mcp-server
  - src/main/kotlin/com/clipping/mcpserver/user/UserSetupSourceController.kt
    - 28:@RequestMapping("/api/user/setup/sources") -> 38: fun searchKnownSources(
      - 38: fun searchKnownSources(
    - 37: @GetMapping("/known-sources") -> 38: fun searchKnownSources(
      - 38: fun searchKnownSources(
    - 61: @GetMapping("/curated") -> 62: fun listCuratedSources(): List {
      - 62: fun listCuratedSources(): List {
    - 78: @PostMapping -> 80: fun create(
      - 80: fun create(
    - 100: @PostMapping("/validate-url") -> 101: fun validateUrl(
      - 101: fun validateUrl(
    - 118: @PostMapping("/{id}/verify") -> 119: fun verify(
      - 119: fun verify(
    - 133: @PostMapping("/{id}/approve") -> 134: fun approve(
      - 134: fun approve(
  - src/main/kotlin/com/clipping/mcpserver/config/SpaController.kt
    - 17: @GetMapping(value = ["/", "/login", "/signup", "/admin/**", "/user/**"], produces = [MediaType.TEXT_HTML_VALUE]) -> 18: fun spa(): Mono = Mono.just(indexHtml)
      - 18: fun spa(): Mono = Mono.just(indexHtml)
  - src/main/kotlin/com/clipping/mcpserver/user/UserDeliveryLogController.kt
    - 16:@RequestMapping("/api/user/delivery-logs") -> 27: fun getDeliveryLogs(
      - 27: fun getDeliveryLogs(
    - 26: @GetMapping -> 27: fun getDeliveryLogs(
      - 27: fun getDeliveryLogs(
  - src/main/kotlin/com/clipping/mcpserver/user/UserArticleHistoryController.kt
    - 22:@RequestMapping("/api/user/history") -> 31: fun searchArticles(
      - 31: fun searchArticles(
    - 30: @GetMapping("/articles") -> 31: fun searchArticles(
      - 31: fun searchArticles(
    - 56: @GetMapping("/articles/{summaryId}") -> 57: fun getArticleDetail(
      - 57: fun getArticleDetail(
    - 70: @PostMapping("/articles/{summaryId}/bookmark") -> 71: fun toggleBookmark(
      - 71: fun toggleBookmark(
  - src/main/kotlin/com/clipping/mcpserver/user/UserClippingRequestController.kt
    - 27:@RequestMapping("/api/user/requests") -> 34: fun list(authentication: Authentication): List =
      - 34: fun list(authentication: Authentication): List =
    - 33: @GetMapping -> 34: fun list(authentication: Authentication): List =
      - 34: fun list(authentication: Authentication): List =
    - 38: @PostMapping -> 40: fun create(
      - 40: fun create(
    - 65: @PostMapping("/{id}/withdraw") -> 66: fun withdraw(
      - 66: fun withdraw(
    - 73: @PostMapping("/{id}/unsubscribe") -> 74: fun unsubscribe(
      - 74: fun unsubscribe(
    - 81: @DeleteMapping("/{id}/remove") -> 83: fun deleteRequest(
      - 83: fun deleteRequest(
    - 94: @PostMapping("/wizard-ownership") -> 96: fun registerWizardOwnership(
      - 96: fun registerWizardOwnership(
    - 116: @PostMapping("/rss-sources") -> 118: fun createAdditionalRssSources(
      - 118: fun createAdditionalRssSources(
  - src/main/kotlin/com/clipping/mcpserver/user/UserDeliveryScheduleController.kt
    - 14:@RequestMapping("/api/user/delivery-schedule") -> 24: fun getSchedule(authentication: Authentication): DeliveryScheduleResponse {
      - 24: fun getSchedule(authentication: Authentication): DeliveryScheduleResponse {
    - 23: @GetMapping -> 24: fun getSchedule(authentication: Authentication): DeliveryScheduleResponse {
      - 24: fun getSchedule(authentication: Authentication): DeliveryScheduleResponse {
    - 33: @PutMapping -> 34: fun updateSchedule(
      - 34: fun updateSchedule(

- src/main/kotlin/com/clipping/mcpserver/user/UserEventController.kt
  - 23:@RequestMapping("/api/user/events") -> 35: fun trackEvents(
    - 35: fun trackEvents(
  - 33: @PostMapping -> 35: fun trackEvents(
    - 35: fun trackEvents(
- src/main/kotlin/com/clipping/mcpserver/user/UserBriefingController.kt
  - 15:@RequestMapping("/api/user/briefing") -> 25: fun getTodayBriefings(
    - 25: fun getTodayBriefings(
  - 24: @GetMapping("/today") -> 25: fun getTodayBriefings(
    - 25: fun getTodayBriefings(
- src/main/kotlin/com/clipping/mcpserver/user/CompanySearchUserController.kt
  - 15:@RequestMapping("/api/user/companies") -> 27: fun search(
    - 27: fun search(
  - 26: @GetMapping -> 27: fun search(
    - 27: fun search(
- src/main/kotlin/com/clipping/mcpserver/user/UserAccountController.kt
  - 17:@RequestMapping("/api/user/account") -> 27: fun selfWithdraw(
    - 27: fun selfWithdraw(
  - 25: @PostMapping("/withdraw") -> 27: fun selfWithdraw(
    - 27: fun selfWithdraw(
- src/main/kotlin/com/clipping/mcpserver/admin/CategoryRuleAdminController.kt
  - 22:@RequestMapping("/api/admin/category-rules") -> 32: fun getStats(@RequestParam(defaultValue = "7") days: Int): RuleStatsResponse {
    - 32: fun getStats(@RequestParam(defaultValue = "7") days: Int): RuleStatsResponse {
  - 31: @GetMapping("/stats") -> 32: fun getStats(@RequestParam(defaultValue = "7") days: Int): RuleStatsResponse {
    - 32: fun getStats(@RequestParam(defaultValue = "7") days: Int): RuleStatsResponse {
  - 54: @GetMapping("/{categoryId}/excluded-items") -> 55: fun getExcludedItems(
    - 55: fun getExcludedItems(
  - 77: @GetMapping("/{categoryId}") -> 78: fun get(@PathVariable categoryId: String): CategoryRuleResponse =
    - 78: fun get(@PathVariable categoryId: String): CategoryRuleResponse =
  - 84: @PutMapping("/{categoryId}") -> 85: fun update(
    - 85: fun update(
- src/main/kotlin/com/clipping/mcpserver/admin/KeywordTrendAdminController.kt
  - 14:@RequestMapping("/api/admin/keywords")
    - 27: fun getKeywordTrend(
  - 26: @GetMapping("/trend") -> 27: fun getKeywordTrend(
    - 27: fun getKeywordTrend(

arc-reactor-admin API 엔드포인트 근거

- 저장소: /Users/jinan/ai/arc-reactor-admin
  - 컨트롤러 구현 파일 미발견(또는 테스트만 존재)

arc-reactor/arc-web API 엔드포인트 근거

- 저장소: /Users/jinan/ai/arc-reactor/arc-web
  - src/main/kotlin/com/arc/reactor/controller/SchedulerController.kt
    - 55:@RequestMapping("/api/scheduler/jobs")
      - 62: @Operation(summary = "전체 예약 작업 목록 조회 (태그 필터 선택)")
      - 68: fun listJobs(
    - 67: @GetMapping -> 68: fun listJobs(
      - 68: fun listJobs(
    - 88: @PostMapping -> 89: fun createJob(
      - 89: fun createJob(
    - 111: @GetMapping("/{id}") -> 112: fun getJob(@PathVariable id: String, exchange: ServerWebExchange): ResponseEntity {
      - 112: fun getJob(@PathVariable id: String, exchange: ServerWebExchange): ResponseEntity {
    - 126: @PutMapping("/{id}") -> 127: fun updateJob(
      - 127: fun updateJob(
    - 150: @DeleteMapping("/{id}") -> 151: fun deleteJob(@PathVariable id: String, exchange: ServerWebExchange): ResponseEntity {
      - 151: fun deleteJob(@PathVariable id: String, exchange: ServerWebExchange): ResponseEntity {
    - 165: @PostMapping("/{id}/trigger") -> 166: fun triggerJob(@PathVariable id: String, exchange: ServerWebExchange): Mono<ResponseEntity> {
      - 166: fun triggerJob(@PathVariable id: String, exchange: ServerWebExchange): Mono<ResponseEntity> {
    - 182: @PostMapping("/{id}/dry-run") -> 183: fun dryRunJob(@PathVariable id: String, exchange: ServerWebExchange): Mono<ResponseEntity> {
      - 183: fun dryRunJob(@PathVariable id: String, exchange: ServerWebExchange): Mono<ResponseEntity> {
    - 199: @GetMapping("/{id}/executions") -> 200: fun getExecutions(
      - 200: fun getExecutions(
  - src/main/kotlin/com/arc/reactor/controller/OutputGuardRuleController.kt
    - 54:@RequestMapping("/api/output-guard/rules")
    - 90: @GetMapping -> 91: fun listRules(exchange: ServerWebExchange): ResponseEntity {
      - 91: fun listRules(exchange: ServerWebExchange): ResponseEntity {
    - 103: @GetMapping("/audits") -> 104: fun listAudits(
      - 104: fun listAudits(
    - 122: @PostMapping -> 123: fun createRule(
      - 123: fun createRule(
    - 165: @PutMapping("/{id}") -> 166: fun updateRule(
      - 166: fun updateRule(
    - 210: @DeleteMapping("/{id}") -> 211: fun deleteRule(
      - 211: fun deleteRule(
    - 236: @PostMapping("/simulate") -> 237: fun simulate(
      - 237: fun simulate(
  - src/main/kotlin/com/arc/reactor/controller/RagIngestionCandidateController.kt
    - 44:@RequestMapping("/api/rag-ingestion/candidates")
      - 57: @Operation(summary = "RAG Ingestion 후보 목록 조회 (관리자)")
    - 62: @GetMapping -> 63: fun list(
      - 63: fun list(
    - 83: @PostMapping("/{id}/approve") -> 84: fun approve(
      - 84: fun approve(
    - 137: @PostMapping("/{id}/reject") -> 138: fun reject(
      - 138: fun reject(

- src/main/kotlin/com/arc/reactor/controller/McpSwaggerCatalogController.kt
  - 49:@RequestMapping("/api/mcp/servers/{name}/swagger/sources")
    - 59: @Operation(summary = "MCP admin API를 통해 Swagger 스펙 소스 목록 조회")
    - 66: suspend fun listSources(
  - 65: @GetMapping -> 66: suspend fun listSources(
    - 66: suspend fun listSources(
  - 87: @GetMapping("/{sourceName}") -> 88: suspend fun getSource(
    - 88: suspend fun getSource(
  - 106: @PostMapping -> 107: suspend fun createSource(
    - 107: suspend fun createSource(
  - 124: @PutMapping("/{sourceName}") -> 125: suspend fun updateSource(
    - 125: suspend fun updateSource(
  - 144: @PostMapping("/{sourceName}/sync") -> 145: suspend fun syncSource(
    - 145: suspend fun syncSource(
  - 163: @GetMapping("/{sourceName}/revisions") -> 164: suspend fun listRevisions(
    - 164: suspend fun listRevisions(
  - 184: @GetMapping("/{sourceName}/diff") -> 185: suspend fun getDiff(
    - 185: suspend fun getDiff(
  - 209: @PostMapping("/{sourceName}/publish") -> 210: suspend fun publishRevision(
    - 210: suspend fun publishRevision(
- src/main/kotlin/com/arc/reactor/controller/ChatController.kt
  - 68:@RequestMapping("/api/chat")
  - 104: @PostMapping -> 105: suspend fun chat(
    - 105: suspend fun chat(
  - 153: @PostMapping("/stream", produces = [MediaType.TEXT_EVENT_STREAM_VALUE]) -> 154: suspend fun chatStream(
    - 154: suspend fun chatStream(
- src/main/kotlin/com/arc/reactor/controller/DocumentController.kt
  - 51:@RequestMapping("/api/documents")
    - 64: @Operation(summary = "벡터 스토어에 문서 추가 (관리자)")
  - 72: @PostMapping -> 73: fun addDocument(
    - 73: fun addDocument(
  - 129: @PostMapping("/batch") -> 130: fun addDocuments(
    - 130: fun addDocuments(
  - 181: @PostMapping("/search") -> 182: fun searchDocuments(@Valid @RequestBody request: SearchDocumentRequest): List {
    - 182: fun searchDocuments(@Valid @RequestBody request: SearchDocumentRequest): List {
  - 213: @DeleteMapping -> 214: fun deleteDocuments(
    - 214: fun deleteDocuments(
- src/main/kotlin/com/arc/reactor/controller/ToolPolicyController.kt
  - 36:@RequestMapping("/api/tool-policy")
    - 49: @Operation(summary = "Tool Policy 상태 조회 (적용 중 + 저장) (관리자)")
  - 54: @GetMapping -> 55: fun get(exchange: ServerWebExchange): ResponseEntity {
    - 55: fun get(exchange: ServerWebExchange): ResponseEntity {
  - 76: @PutMapping -> 77: fun update(
    - 77: fun update(
  - 104: @DeleteMapping -> 105: fun delete(exchange: ServerWebExchange): ResponseEntity {
    - 105: fun delete(exchange: ServerWebExchange): ResponseEntity {
- src/main/kotlin/com/arc/reactor/controller/GlobalExceptionHandler.kt
  - (매핑 애노테이션 미발견)
- src/main/kotlin/com/arc/reactor/controller/RagIngestionPolicyController.kt
  - 37:@RequestMapping("/api/rag-ingestion/policy")
    - 50: @Operation(summary = "RAG Ingestion Policy 상태 조회 (적용 중 + 저장) (관리자)")
  - 55: @GetMapping -> 56: fun get(exchange: ServerWebExchange): ResponseEntity {
    - 56: fun get(exchange: ServerWebExchange): ResponseEntity {
  - 75: @PutMapping -> 76: fun update(
    - 76: fun update(
  - 103: @DeleteMapping -> 104: fun delete(exchange: ServerWebExchange): ResponseEntity {
    - 104: fun delete(exchange: ServerWebExchange): ResponseEntity {
- src/main/kotlin/com/arc/reactor/controller/McpPreflightController.kt
  - 37:@RequestMapping("/api/mcp/servers/{name}/preflight")
    - 47: @Operation(summary = "MCP 서버 admin preflight 점검 실행 (프록시)")
  - 55: @GetMapping -> 56: suspend fun getPreflight(
    - 56: suspend fun getPreflight(

- src/main/kotlin/com/arc/reactor/controller/PromptLabController.kt
  - 63:@RequestMapping("/api/prompt-lab")
  - 94: @PostMapping("/experiments") -> 95: fun createExperiment(
    - 95: fun createExperiment(
  - 129: @GetMapping("/experiments") -> 130: fun listExperiments(
    - 130: fun listExperiments(
  - 147: @GetMapping("/experiments/{id}") -> 148: fun getExperiment(
    - 148: fun getExperiment(
  - 167: @PostMapping("/experiments/{id}/run") -> 168: fun runExperiment(
    - 168: fun runExperiment(
  - 208: @PostMapping("/experiments/{id}/cancel") -> 209: fun cancelExperiment(
    - 209: fun cancelExperiment(
  - 236: @GetMapping("/experiments/{id}/status") -> 237: fun getStatus(
    - 237: fun getStatus(
  - 261: @GetMapping("/experiments/{id}/trials") -> 262: fun getTrials(
    - 262: fun getTrials(
  - 278: @GetMapping("/experiments/{id}/report") -> 279: fun getReport(
    - 279: fun getReport(
  - 295: @DeleteMapping("/experiments/{id}") -> 296: fun deleteExperiment(
    - 296: fun deleteExperiment(
  - 314: @PostMapping("/auto-optimize") -> 315: fun autoOptimize(
    - 315: fun autoOptimize(
  - 353: @PostMapping("/analyze") -> 354: suspend fun analyzeFeedback(
    - 354: suspend fun analyzeFeedback(
  - 374: @PostMapping("/experiments/{id}/activate") -> 375: fun activateRecommended(
    - 375: fun activateRecommended(
- src/main/kotlin/com/arc/reactor/controller/MultipartChatController.kt
  - 38:@RequestMapping("/api/chat")
  - 67: @PostMapping("/multipart", consumes = [MediaType.MULTIPART_FORM_DATA_VALUE]) -> 68: suspend fun chatMultipart(
    - 68: suspend fun chatMultipart(

arc-reactor/arc-slack API 엔드포인트 근거

- 저장소: /Users/jinan/ai/arc-reactor/arc-slack
  - src/main/kotlin/com/arc/reactor/slack/controller/SlackEventController.kt
    - 35:@RequestMapping("/api/slack")
      - 39:@Tag(name = "Slack", description = "Slack 이벤트 웹훅 엔드포인트")
      - 45: @Operation(summary = "Slack 이벤트 콜백 수신 (URL 검증 포함)")
      - 50: suspend fun handleEvent(
    - 44: @PostMapping("/events") -> 50: suspend fun handleEvent(
      - 45: @Operation(summary = "Slack 이벤트 콜백 수신 (URL 검증 포함)")
      - 50: suspend fun handleEvent(
  - src/main/kotlin/com/arc/reactor/slack/controller/SlackCommandController.kt
    - 30:@RequestMapping("/api/slack")
      - 34:@Tag(name = "Slack", description = "Slack 슬래시 명령 처리 엔드포인트")
    - 49: @PostMapping("/commands", consumes = [MediaType.APPLICATION_FORM_URLENCODED_VALUE]) -> 55: fun handleSlashCommandEndpoint(
      - 50: @Operation(summary = "Slack 슬래시 명령 수신 및 처리")
      - 55: fun handleSlashCommandEndpoint(

arc-reactor-web API 엔드포인트 근거

- 저장소: /Users/jinan/ai/arc-reactor-web
  - 컨트롤러 구현 파일 미발견(또는 테스트만 존재)

Aslan-Verse-Web API 엔드포인트 근거

- 저장소: /Users/jinan/ai/Aslan-Verse-Web
  - 컨트롤러 구현 파일 미발견(또는 테스트만 존재)

8) 이번 주기 변경량 요약

arc-reactor 변경량 요약

- 최근 2분 커밋: 없음
- 현재 워크트리 변경(최대 8):
  - ?? bin/
  - ?? playwright_out.pdf
  - ?? reports/
  - ?? scripts/ops/generate-arc-reactor-docs-pdf.sh
  - ?? scripts/ops/run-pdfs.sh

aslan-iam 변경량 요약

- 최근 2분 커밋: 없음
- 현재 워크트리 변경(최대 8):
  - M docker-compose.yml
  - M iam-admin-api/src/main/resources/application.yml
  - M iam-api/src/main/resources/application.yml

swagger-mcp-server 변경량 요약

- 최근 2분 커밋: 없음
- 현재 워크트리 변경: 없음

atlassian-mcp-server 변경량 요약

- 최근 2분 커밋: 없음
- 현재 워크트리 변경: 없음

clipping-mcp-server 변경량 요약

- 최근 2분 커밋: 없음
- 현재 워크트리 변경: 없음

arc-reactor-admin 변경량 요약

- 최근 2분 커밋: 없음
- 현재 워크트리 변경: 없음

arc-reactor/arc-web 변경량 요약

- 최근 2분 커밋: 없음
- 현재 워크트리 변경(최대 8):
  - ?? ../bin/
  - ?? ../playwright_out.pdf
  - ?? ../reports/
  - ?? ../scripts/ops/generate-arc-reactor-docs-pdf.sh
  - ?? ../scripts/ops/run-pdfs.sh

arc-reactor/arc-slack 변경량 요약

- 최근 2분 커밋: 없음
- 현재 워크트리 변경(최대 8):
  - ?? ../bin/
  - ?? ../playwright_out.pdf
  - ?? ../reports/
  - ?? ../scripts/ops/generate-arc-reactor-docs-pdf.sh
  - ?? ../scripts/ops/run-pdfs.sh

arc-reactor-web 변경량 요약

- 최근 2분 커밋: 없음
- 현재 워크트리 변경: 없음

Aslan-Verse-Web 변경량 요약

- 최근 2분 커밋: 없음
- 현재 워크트리 변경: 없음

9) 저장소 상호연동 근거(키워드 매칭)

- 분석 방식: 각 저장소의 구성 파일/컨트롤러에서 다른 컴포넌트 명칭을 근거로 상호 연동 단서를 추출합니다.

arc-reactor

- 저장소: /Users/jinan/ai/arc-reactor
  - aslan-iam 언급/연동 증거
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:12:- 분석 범위: /Users/jinan/ai 내 아슬란 생태계 프로젝트 8개 (arc-reactor, aslan-iam, swagger-mcp-server, atlassian-mcp-server, clipping-mcp-server, arc-reactor-admin, arc-reactor-web, Aslan-Verse-Web)
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:33:- Arc Reactor는 aslan-iam 인증 기반 + MCP 생태계 동적 등록 + 웹/슬랙 채널 확장 + 관리 콘솔로 이어지는 중앙형 플랫폼입니다.
  - swagger-mcp-server 언급/연동 증거
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:12:- 분석 범위: /Users/jinan/ai 내 아슬란 생태계 프로젝트 8개 (arc-reactor, aslan-iam, swagger-mcp-server, atlassian-mcp-server, clipping-mcp-server, arc-reactor-admin, arc-reactor-web, Aslan-Verse-Web)
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:42:| swagger-mcp-server | /Users/jinan/ai/swagger-mcp-server | API 도구 MCP | OK | /Users/jinan/ai/swagger-mcp-server/README.md | Swagger MCP Server | main | 8964374 | clean | 2 weeks ago |
  - atlassian-mcp-server 언급/연동 증거
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:12:- 분석 범위: /Users/jinan/ai 내 아슬란 생태계 프로젝트 8개 (arc-reactor, aslan-iam, swagger-mcp-server, atlassian-mcp-server, clipping-mcp-server, arc-reactor-admin, arc-reactor-web, Aslan-Verse-Web)
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:43:| atlassian-mcp-server | /Users/jinan/ai/atlassian-mcp-server | Jira/Confluence/Bitbucket MCP | OK | /Users/jinan/ai/atlassian-mcp-server/README.md | atlassian-mcp-server | main | 054a127 | clean | 2 weeks ago |
  - clipping-mcp-server 언급/연동 증거
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:12:- 분석 범위: /Users/jinan/ai 내 아슬란 생태계 프로젝트 8개 (arc-reactor, aslan-iam, swagger-mcp-server, atlassian-mcp-server, clipping-mcp-server, arc-reactor-admin, arc-reactor-web, Aslan-Verse-Web)
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:44:| clipping-mcp-server | /Users/jinan/ai/clipping-mcp-server | 클리핑·요약 MCP | OK | /Users/jinan/ai/clipping-mcp-server/README.md | clipping-mcp-server | main | 23abc52 | clean | 3 hours ago |
  - arc-reactor-admin 언급/연동 증거
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:12:- 분석 범위: /Users/jinan/ai 내 아슬란 생태계 프로젝트 8개 (arc-reactor, aslan-iam, swagger-mcp-server, atlassian-mcp-server, clipping-mcp-server, arc-reactor-admin, arc-reactor-web, Aslan-Verse-Web)
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:45:| arc-reactor-admin | /Users/jinan/ai/arc-reactor-admin | 운영/모니터링 UI | OK | /Users/jinan/ai/arc-reactor-admin/README.md | arc-reactor-admin | main | ca42355 | clean | 3 hours ago |
  - arc-reactor-web 언급/연동 증거
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:12:- 분석 범위: /Users/jinan/ai 내 아슬란 생태계 프로젝트 8개 (arc-reactor, aslan-iam, swagger-mcp-server, atlassian-mcp-server, clipping-mcp-server, arc-reactor-admin, arc-reactor-web, Aslan-Verse-Web)
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:48:| arc-reactor-web | /Users/jinan/ai/arc-reactor-web | 사용자 UI | OK | /Users/jinan/ai/arc-reactor-web/README.ko.md | Arc Reactor Web | main | e0d84a8 | clean | 3 hours ago |
  - MCP 언급/연동 증거
    - /Users/jinan/ai/arc-reactor/KNOWN_ACCEPTABLE.md:84:| `McpAdminHmacSupport.kt:64` hex + MessageDigest per request | P4 | MCP admin proxy 전용, 저빈도 | 2026-03-13 |
    - /Users/jinan/ai/arc-reactor/KNOWN_ACCEPTABLE.md:97:| `McpServerController.kt:432` getStatus per server in list | P4 | MCP 서버 수 소규모 | 2026-03-13 |
  - JWT 언급/연동 증거
    - /Users/jinan/ai/arc-reactor/KNOWN_ACCEPTABLE.md:215:| `JwtAuthWebFilter.kt:57` blank sub → "anonymous" | P4 | JWT 서명키 탈취 전제. 키 보유 시 어떤 userId도 위장 가능. 방어적 폴백 수준 | 2026-03-14 |
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:53:- Arc Reactor의 인증 중심은 aslan-iam의 공개키 검증 흐름과 결합되어 JWT를 로컬 검증하고, 로그인·권한·토큰 발급은 aslan-iam에서 분리 운영합니다 (`aslan-iam/README.md`, `aslan-iam/README.md`의 API 항목).
  - SSE 언급/연동 증거
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:21:- 용어 정합성: Arc Reactor, MCP, IAM, ReAct, Slack, SSE를 일관된 표기 사용
    - /Users/jinan/ai/arc-reactor/reports/arc-reactor-full-feature-guide.md:56:- 웹 채널(`arc-reactor-web`)은 세션 채팅, SSE 스트리밍, Persona/Tool Policy/Output Guard/스케줄러 관리 화면을 제공합니다 (`arc-reactor-web/README.ko.md`).
  - tenant 언급/연동 증거
    - /Users/jinan/ai/arc-reactor/KNOWN_ACCEPTABLE.md:24:| `TenantResolver.kt:83` (manager tenant override) | P3 | Self-hosted 배포 설계 의도 (기존 항목과 동일) | 2026-03-12 |
    - /Users/jinan/ai/arc-reactor/KNOWN_ACCEPTABLE.md:95:| `PlatformAdminController.kt:279` tenantAnalytics N+1 | P4 | 관리자 대시보드, 저빈도 | 2026-03-13 |

aslan-iam

- 저장소: /Users/jinan/ai/aslan-iam
  - JWT 언급/연동 증거
    - /Users/jinan/ai/aslan-iam/CLAUDE.md:21: ├── iam-infra/ # JWT, TOTP, BCrypt 어댑터 (순수 어댑터만)
    - /Users/jinan/ai/aslan-iam/CLAUDE.md:28: ├── iam-infra (JWT/TOTP/BCrypt 어댑터) │

swagger-mcp-server

- 저장소: /Users/jinan/ai/swagger-mcp-server
  - MCP 언급/연동 증거
    - /Users/jinan/ai/swagger-mcp-server/src/main/kotlin/com/swagger/mcpserver/controller/AdminPreflightController.kt:15:@Tag(name = "Admin", description = "Admin endpoints (requires SWAGGER_MCP_ADMIN_TOKEN header)")
    - /Users/jinan/ai/swagger-mcp-server/src/main/kotlin/com/swagger/mcpserver/controller/AdminPreflightController.kt:25: @Operation(summary = "Run production-readiness preflight checks for swagger MCP")
  - JWT 언급/연동 증거
    - /Users/jinan/ai/swagger-mcp-server/README.md:233:curl -X POST -H "Authorization: Bearer $JWT" -H "Content-Type: application/json" \
  - SSE 언급/연동 증거
    - /Users/jinan/ai/swagger-mcp-server/README.md:20:# SSE 서버 실행 (기본, 포트 8081)
    - /Users/jinan/ai/swagger-mcp-server/README.md:102: | MCP | spring-ai-starter-mcp-server-webflux (SSE + STDIO) |

atlassian-mcp-server

- 저장소: /Users/jinan/ai/atlassian-mcp-server
  - MCP 언급/연동 증거
    - /Users/jinan/ai/atlassian-mcp-server/src/main/kotlin/com/atlassian/mcpserver/tool/PersonalRequesterIdentityResolver.kt:84: "MCP_REQUIRE_EXPLICIT_ASSIGNEE_FOR_PERSONAL_TOOLS=true"
    - /Users/jinan/ai/atlassian-mcp-server/src/main/kotlin/com/atlassian/mcpserver/model/JiraModels.kt:136:// ===== Simplified Info for MCP Tool Responses =====
  - SSE 언급/연동 증거
    - /Users/jinan/ai/atlassian-mcp-server/build.gradle.kts:29: // Spring AI MCP Server (SSE via Servlet/WebMVC)
    - /Users/jinan/ai/atlassian-mcp-server/README.md:381:Server starts on port **8085** with SSE transport.
  - tenant 언급/연동 증거
    - /Users/jinan/ai/atlassian-mcp-server/README.md:192: | `ATLASSIAN_BASE_URL` | Yes | Your Atlassian site URL (browser URL), e.g. `https://<site>.atlassian.net` | `curl -s "$ATLASSIAN_BASE_URL/_edge/tenant_info"` |
    - /Users/jinan/ai/atlassian-mcp-server/README.md:196: | `ATLASSIAN_CLOUD_ID` | Required when gateway mode is enabled | `curl -s "$ATLASSIAN_BASE_URL/_edge/tenant_info" \| jq -r '.cloudId'` | Must be non-empty UUID |

clipping-mcp-server

- 저장소: /Users/jinan/ai/clipping-mcp-server
  - MCP 언급/연동 증거
    - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/tool/ToolResponse.kt:16: * MCP 툴 실행 시 서비스 계층 예외를 일관된 JSON 에러 응답으로 변환합니다.
    - /Users/jinan/ai/clipping-mcp-server/build.gradle.kts:38: // Spring AI MCP Server (SSE via WebFlux)
  - JWT 언급/연동 증거
    - /Users/jinan/ai/clipping-mcp-server/docs/superpowers/specs/2026-03-19-e2e-test-suite-design.md:622: | cross-flow #10 (탈퇴→로그인 불가) | JWT 세션이면 즉시 무효화 안 될 수 있음 | 백엔드 세션 방식 확인 후 조정 |
    - /Users/jinan/ai/clipping-mcp-server/docs/CLIPPING_GAP_ANALYSIS.md:46:- Add API authn/authz for `/api/admin/**` (JWT or service token + role check)
  - SSE 언급/연동 증거
    - /Users/jinan/ai/clipping-mcp-server/build.gradle.kts:38: // Spring AI MCP Server (SSE via WebFlux)
    - /Users/jinan/ai/clipping-mcp-server/src/main/kotlin/com/clipping/mcpserver/admin/AdminUserRequestController.kt:137: is com.clipping.mcpserver.error.InvalidInputException -> "ALREADY_PROCESSED"
  - tenant 언급/연동 증거
    - /Users/jinan/ai/clipping-mcp-server/docs/CLIPPING_GAP_ANALYSIS.md:34:4. Single-tenant policy clarity gap (Low)
    - /Users/jinan/ai/clipping-mcp-server/docs/CLIPPING_GAP_ANALYSIS.md:35: - This repository is currently operated for a single company tenant.

arc-reactor-admin

- 저장소: /Users/jinan/ai/arc-reactor-admin
  - swagger-mcp-server 언급/연동 증거
    - /Users/jinan/ai/arc-reactor-admin/README.md:11:로컬 operator stack(`arc-reactor` + `swagger-mcp-server` + `atlassian-mcp-server` + admin dev server)이 떠 있는 상태라면:
  - atlassian-mcp-server 언급/연동 증거
    - /Users/jinan/ai/arc-reactor-admin/README.md:11:로컬 operator stack(`arc-reactor` + `swagger-mcp-server` + `atlassian-mcp-server` + admin dev server)이 떠 있는 상태라면:
  - MCP 언급/연동 증거
    - /Users/jinan/ai/arc-reactor-admin/README.md:28:이 스크립트는 기본적으로 `http://127.0.0.1:18081`에 로그인해서 actuator, capabilities, MCP registry/preflight/policy,
    - /Users/jinan/ai/arc-reactor-admin/README.md:29:MCP security, tool policy, scheduler, approvals, audit, output guard 계약을 확인합니다.
  - JWT 언급/연동 증거
    - /Users/jinan/ai/arc-reactor-admin/docs/superpowers/specs/2026-03-21-ux-improvements-design.md:310:- JWT `exp` parsing utility in `shared/lib/jwt.ts`: manual base64url decode (`atob`) + `JSON.parse` to extract payload. No external library needed. Returns `exp` as Unix timestamp or `null` if token is not a valid JWT.
    - /Users/jinan/ai/arc-reactor-admin/docs/superpowers/specs/2026-03-21-ux-improvements-design.md:338:- **Session expiry**: Unit test for JWT parsing utility, integration test for timer behavior
  - SSE 언급/연동 증거
    - /Users/jinan/ai/arc-reactor-admin/pnpm-lock.yaml:1997: resolution: {integrity: sha512-1to4zXBxmXHV3IiSSEInrreIlu02vUOvrhxJJH5vcxYTBDAx51cqZiKdyTxlecdKNSjj8EcxGBxNf6Vg+945gw==}
    - /Users/jinan/ai/arc-reactor-admin/docs/admin-qa-checklist.md:32:2. Confirm transport help text explains `SSE`, `STDIO`, `STREAMABLE_HTTP`.
  - tenant 언급/연동 증거
    - /Users/jinan/ai/arc-reactor-admin/docs/admin-operator-roadmap.md:18:- Platform-level and tenant-level admin surfaces
    - /Users/jinan/ai/arc-reactor-admin/docs/admin-operator-roadmap.md:100:- Multi-tenant bulk administration and export workflows

arc-reactor-web

- 저장소: /Users/jinan/ai/arc-reactor-web
  - arc-reactor-admin 언급/연동 증거
    - /Users/jinan/ai/arc-reactor-web/docs/prd-admin-features.md:577:Admin 기능을 독립적인 Vite 앱(`arc-reactor-admin`)으로 분리한다.
  - MCP 언급/연동 증거
    - /Users/jinan/ai/arc-reactor-web/README.ko.md:34:- 어드민 대시보드 — MCP 서버, 페르소나, 인텐트, 아웃풋 가드, 툴 정책, 스케줄러, 클리핑
    - /Users/jinan/ai/arc-reactor-web/README.ko.md:44:| `/admin/mcp-servers` | MCP 서버 — 등록 및 연결 관리 |
  - JWT 언급/연동 증거
    - /Users/jinan/ai/arc-reactor-web/CLAUDE.md:161:JWT stored in `localStorage` as `arc-reactor-auth-token`. The `api` ky instance injects it via `beforeRequest` hook and clears it on 401. Roles: `USER`, `ADMIN`.
    - /Users/jinan/ai/arc-reactor-web/README.ko.md:31:- 선택적 JWT 인증 및 사용자 격리
  - SSE 언급/연동 증거
    - /Users/jinan/ai/arc-reactor-web/CLAUDE.md:107:Use the `api` instance from `src/lib/http.ts` in service files. Do not use `fetch` directly (except `streamChat` which uses SSE).
    - /Users/jinan/ai/arc-reactor-web/README.ko.md:26:- SSE(Server-Sent Events) 기반 실시간 스트리밍 응답

Aslan-Verse-Web

- 저장소: /Users/jinan/ai/Aslan-Verse-Web
  - JWT 언급/연동 증거
    - /Users/jinan/ai/Aslan-Verse-Web/docs/superpowers/specs/2026-04-05-aslan-verse-web-fe-design.md:367:- JWT Bearer token 방식인지 확인 필요
  - SSE 언급/연동 증거
    - /Users/jinan/ai/Aslan-Verse-Web/docs/superpowers/specs/2026-04-05-aslan-verse-web-fe-design.md:31:| 실시간 스트리밍 | Native `EventSource` (SSE) | - |
    - /Users/jinan/ai/Aslan-Verse-Web/docs/superpowers/specs/2026-04-05-aslan-verse-web-fe-design.md:38:**SSE vs axios 결정:** ky를 일반 REST 호출에 사용하고, SSE는 native `EventSource`로 분리 처리. axios 미사용.

1) 한눈에 보는 개요

Arc Reactor는 Spring AI 기반 AI Agent 런타임의 핵심을 제공하는 엔터프라이즈 플랫폼입니다.
`arc-core`를 중심으로 `ReAct` 실행, 보안 `Guard`, 확장성 `Hook`, 동적 `MCP` 연결, 스케줄러를 한 곳에서 운영할 수 있으며, 웹/Slack/관리 UI까지 동일 정책으로 통합됩니다.

- 핵심 런타임: `arc-reactor` (`arc-core`, `arc-admin`, 실행 조립 `arc-app`)
- 채널: `arc-reactor-web`(웹 UI), `arc-slack`(Slack Events/SocketMode), REST API

- 운영 플랫폼: `arc-reactor-admin`(대시보드/운영 승인/감사)
- 권한/인증: `aslan-iam`(외부 인증 발급, 공개키 기반 토큰 검증)
- 도구(도메인 MCP): `clipping-mcp-server`, `swagger-mcp-server`, `atlassian-mcp-server`
- 실험 인프라: `Aslan-Verse-Web`(멀티 리액터 조직·페르소나 실험)

이 설계는 "한 팀의 채팅봇"이 아니라 "회사 단위 AI 오케스트레이션 플랫폼"으로 보는 게 정확합니다.

2) 플랫폼 맵 (전체 시스템 연결)

2.1 요청 흐름

1. 사용자/시스템 요청 수신: `arc-reactor-web` 또는 `arc-slack` 또는 직접 API
2. `Guard` 단계: 입력 검증, 길이 제한, 보안 룰, 레이트리밋/인증 검사
3. `Hook(BeforeStart)`: 운영 메타 주입(테넌트/세션/추적값)
4. `ReAct Loop`: LLM ↔ Tool 호출 반복
5. `Hook(AfterToolCall/AfterComplete/AfterResponse)` 처리
6. 응답 반환 및 감사 로그 저장

2.2 인증/권한 경로

- `aslan-iam`에서 JWT 공개키를 받아 토큰의 서명 검증을 수행하고, `arc-reactor` 및 연동 MCP가 네트워크 검증을 최소화합니다.
- 토큰이 없는 요청은 `anonymous` 정책 폴백 처리.
- 운영 정책은 채널별(웹/슬랙/관리), 테넌트별, 도메인별로 구분합니다.

2.3 MCP 연동 구조

- `arc-reactor`의 REST API를 통해 MCP 서버를 동적으로 등록·수정·삭제
- 서버 타입/프로토콜 기반 동적 라우팅(REST/SSE/STDIO 패턴)
- 프라이빗 주소 접근/토큰/HMAC 등 네트워크 경계 정책 적용

3) 핵심 런타임 기능 (`arc-core`)

3.1 ReAct 엔진

- 채팅 기반 추론/행동 루프
- `maxToolCalls`, `maxToolsPerRequest`, tool 타임아웃, loop timeout으로 무한 루프 방지
- Assist/Tool 응답 쌍의 무결성 요구(도구 호출 추적 정합성)

3.2 Guard (Fail-Close)

- 기본 5단계 Guard 집합: 입력 정규화, 제로폭/이상문자 검사, 경계값 검사, 보안/인젝션 규칙, 승인 정책
- 승인/차단 실패 시 즉시 중단
- 정책 위반을 명시적 에러 응답으로 반환

3.3 Hook (Fail-Open)

- `BeforeStart`, `AfterToolCall`, `BeforeResponse`, `AfterComplete`
- Hook 실패는 요청 중단 원인으로 전파하지 않고 로깅/알림 중심으로 격리
- 운영 모니터링과 비용 회수 지표 수집에 유리

3.4 Safety, Policy, Output Guard

- Tool 출력 정제(`ToolOutputSanitizer`), PII 마스킹, 출력 가드 규칙
- 권한이 과한 텍스트/민감 데이터 유출 패턴을 규칙 기반으로 차단

3.5 HITL(Human in the loop)

- 위험한 Tool은 승인 큐로 라우팅 가능
- 승인자/타임아웃/재요청 정책 지정
- 관리자 승인/감사 체계와 결합 시 규정 준수성 강화

3.6 Scheduler

- Spring cron 기반 주기적 작업 등록
- MCP 작업, 내부 에이전트 작업, 메시지 알림/보고 자동화 지원
- 재시도 및 실패 핸들링 옵션 제공

3.7 멀티 에이전트 실험성

- 테스트용 Supervisor/Prompt Variant/평가 루프 기반 실험이 가능
- 채팅템플릿, 페르소나, 도구 접근권한을 버전별로 운영

4) 채널별 기능

4.1 웹 채널 (`arc-reactor-web`)

- 세션 대화, 스트리밍 채팅, 첨부 업로드, 세션 내보내기 지원
- API 통합을 통한 기능 토글: persona, promptTemplate, outputGuard, mcpServers, scheduler, rate policy
- 다국어 UI 및 반응형 레이아웃

4.2 Slack 채널 (`arc-slack`)

- Socket Mode와 Events API 모두 대응
- 이벤트/멘션/DM, 슬래시 명령 처리
- 쓰레드 기반 사용자 세션 매핑
- Slack 내 중복 처리 억제 및 서명 검증

4.3 관리자 채널 (`arc-reactor-admin`)

- 운영 지표: MCP 상태, 요청/도구 실패율, 로그 연동 상태
- 정책 운영: 허용 도구, 금지 규칙, outputGuard 및 승인 정책
- 관리자 승인 흐름 및 감사 추적

5) MCP 서버 상세

5.1 aslan-iam

- 조직 사용자 발급/관리, JWT 공개키 기반 검증
- arc-reactor 계열 서비스에서 사용자 및 역할 판단의 단일 권한 소스 역할

5.2 swagger-mcp-server

- OpenAPI 명세 탐색/요약/검증
- API 카탈로그 동기화 및 spec 운영 업무 자동화

5.3 atlassian-mcp-server

- Jira, Confluence, Bitbucket 도구군 중심으로 작업, 문서, 코드 이력까지 연결
- 프로젝트/공간 단위 권한 분리

5.4 clipping-mcp-server

- RSS/뉴스/자료 수집, 요약 생성, 다이제스트 정기 발송
- 팀별/페르소나별 정보 축적용으로 실무 브리핑 자동화

6) 실험형 확장: Aslan-Verse-Web

요청하신 실험 시나리오("여러 팀의 가상 페르소나 조직" 구성) 관점에서 핵심입니다.

- 팀 단위 조직 템플릿(Workspace/Persona/Task/Network)을 구성해 ReAct 실행 정책을 분리 운영
- 같은 회사/팀 계열도 페르소나를 달리해 다른 의사결정 패턴을 병렬 테스트
- Arc Reactor의 Persona/Prompt 정책과 결합해 시뮬레이션 결과를 Slack/Web/Admin 채널로 동기화 가능

7) 운영 관점 체크리스트

- 비밀키가 하드코딩되어 있지 않은지 점검 (`jwt-secret` 등)
- Guard/Hook 동작 모드와 승인 규칙을 단계적으로 강화
- MCP private network 제한 정책 적용
- 스케줄러 빈도·실행시간·비용(토큰) 한도 정책
- 감사 로그 보존 기간과 검색성 설정
- 배포 전 hardening 테스트(적대 입력/정상 입력 동시)

8) 도입 우선순위(권장)

1. 아키텍처 고정: `arc-reactor` + `aslan-iam` 연동
2. 운영 채널 연결: `arc-reactor-web`, `arc-reactor-admin`, `arc-slack`
3. 도구 확장: `swagger`, `atlassian`, `clipping` MCP 등록
4. 정책 수립: outputGuard, HITL, rate-limit, hook/guard 정책
5. 확장 실험: `Aslan-Verse-Web` 페르소나 조직으로 멀티 리액터 운영 검증

9) 문서화/공유 포맷

- 이 문서(MD): 기능 카탈로그 기준
- 실행 결과 PDF: 제품 소개 및 전사 공유용
- 다음 문서: CTO/임원용 활용 전략 문서(사용 시나리오 + 기대효과 + 와우 포인트)